

Exhibits: figures and tables

Who Benefits from Food Tax Exemptions in Lower-Income Settings? Evidence on Pass-Through and Incidence

Figures and tables in paper order. Figure titles and notes match the paper; each figure is the file produced by the replication package (see `docs/exhibit_manifest.md`).

Main tables and figures

Table 1: Summary of data sources

	Supermarket census	Administrative data	Scraping of websites	Phone survey panel
Location	Port Moresby	Port Moresby	Port Moresby	Nationwide
Unit	Store	Store	Store	Household
Store type	Formal	Formal	Formal	Formal + Informal
Coverage	All	Subset	All online	Representative
Frequency	Monthly	Monthly	Weekly	Monthly
Exempt items	10	10	10	8
Non-exempt items	10	129	479	8
Time period	May-Nov 25	Dec 23 - May 26	May-Aug 25	Oct 23 - May 26
Strengths	All formal stores	Multi-year	All products	Consumption data
Limitations	Limited pre-trends	Non-representative	Only two chains	Self-reported

Notes: This table summarizes the four data sources used in the analysis. Location refers to geographic scope. Unit refers to whether the data covers stores or households. Store type refers to whether the data sources cover just formal or formal and informal stores. Coverage refers to whether the data source captures the universe of relevant outlets or a subset. Frequency refers to how often the data was collected. “Exempt items” and “non-exempt items” denote the number of treated and comparison-group product categories used in each source. Time period refers to the period for which this data source was collected. In the case of the scraped website data, there are 97 specific products within the broader 10 exempt food item categories.

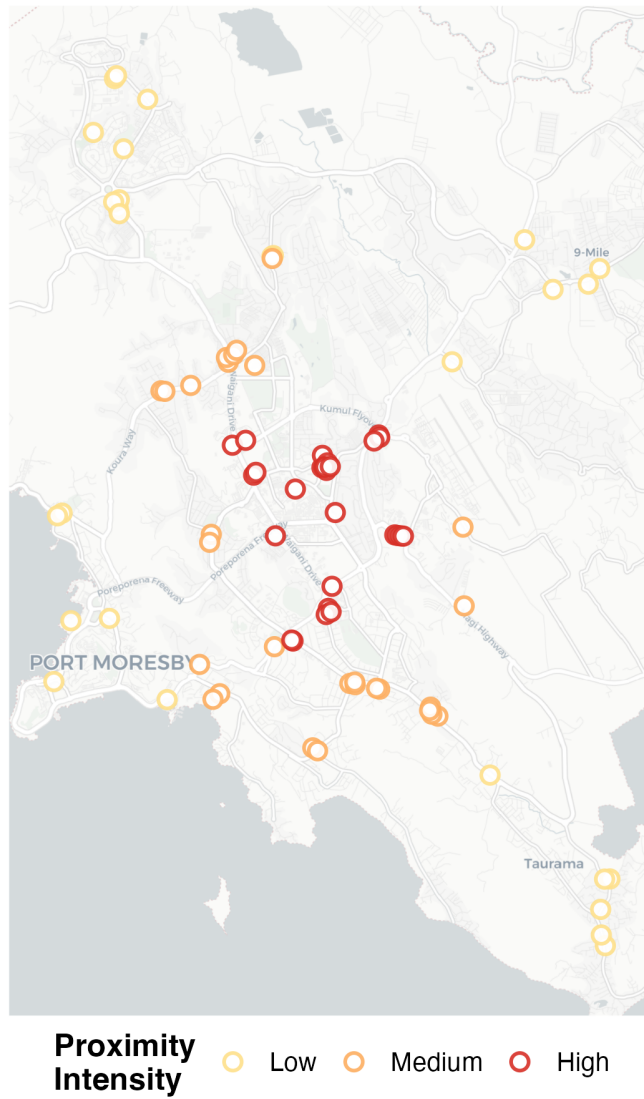


Figure 1: Map of Port Moresby with supermarket locations

Notes: This figure maps the 83 formal supermarkets included in the Port Moresby supermarket census. Marker shading indicates stores' proximity to competitors, based on the Ahlfeldt–Redding–Sturm–Wolf proximity measure, with darker markers denoting greater local competitive intensity. High-proximity stores are concentrated in central Port Moresby.

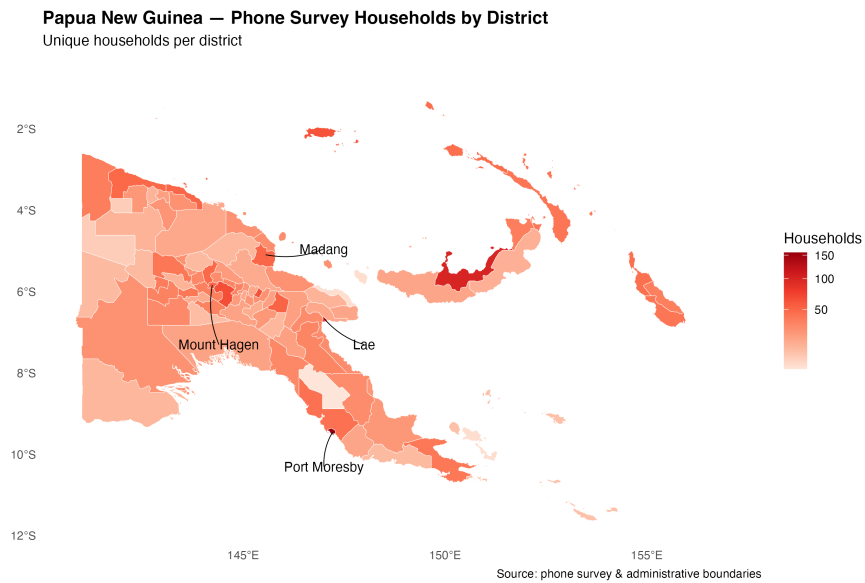
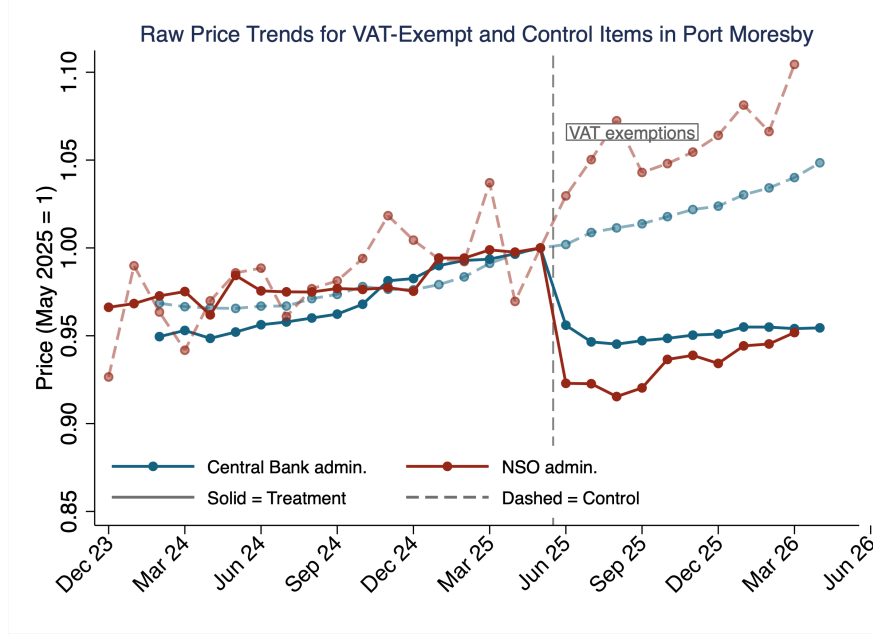


Figure 2: Map of Papua New Guinea with respondents' locations

Notes: This figure maps the geographic distribution of households in the monthly phone survey panel. Shading indicates the number of unique respondent households observed in each district over the sample period. The figure illustrates the survey's broad national coverage.

(a) Raw trends



(b) Differences-in-Differences results

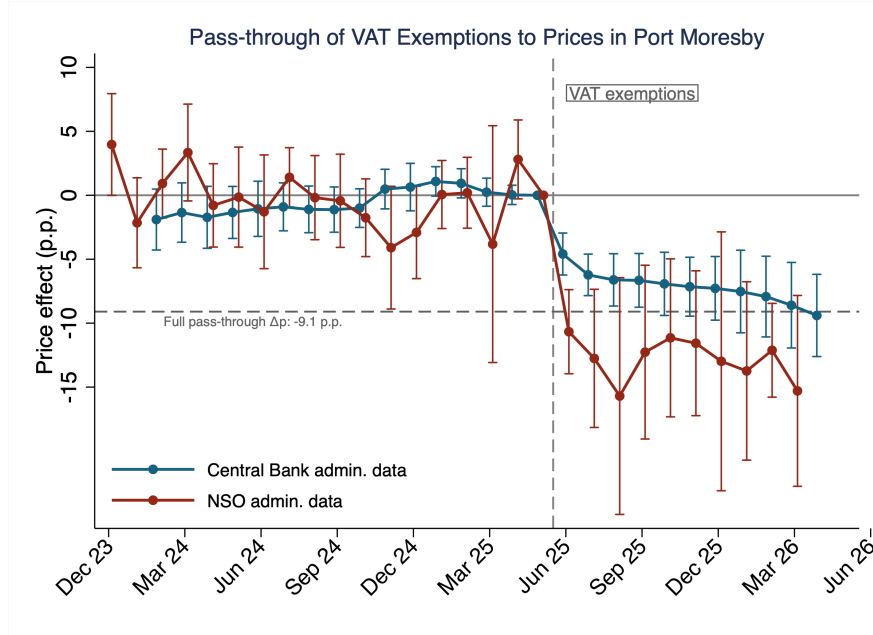
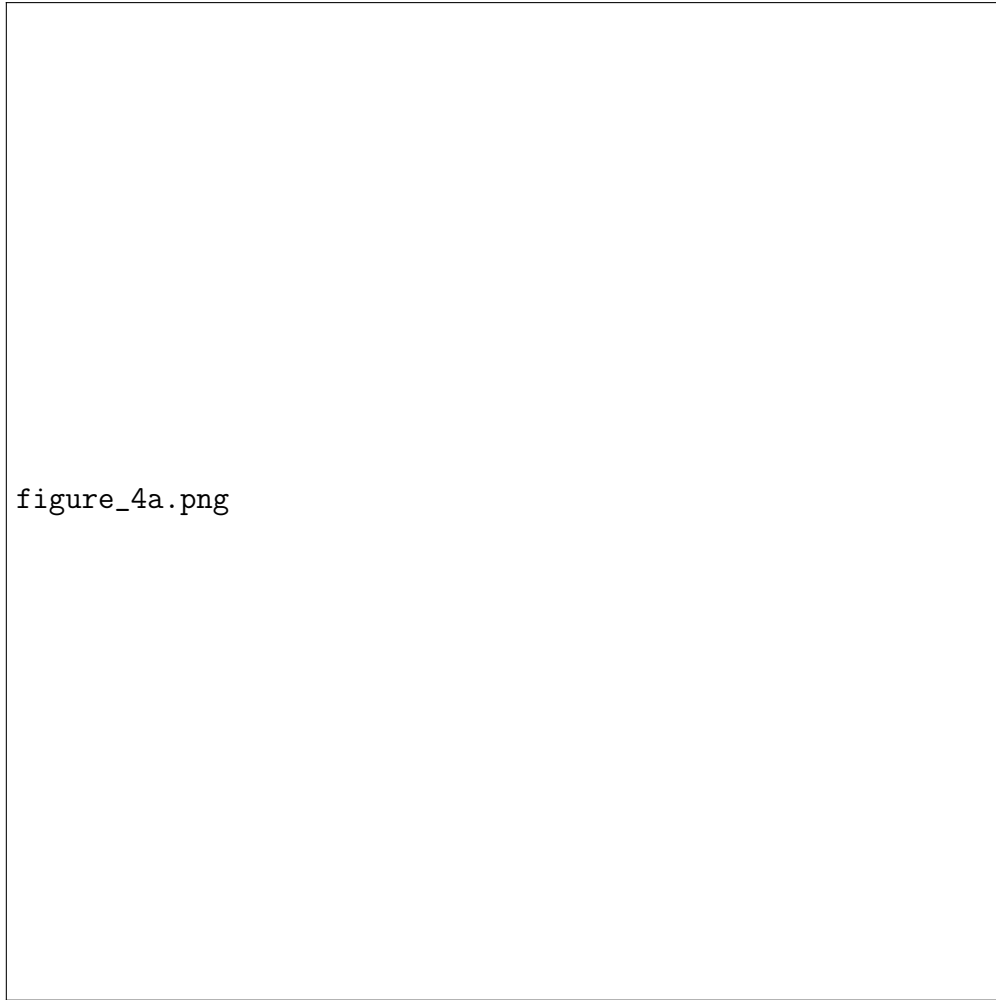


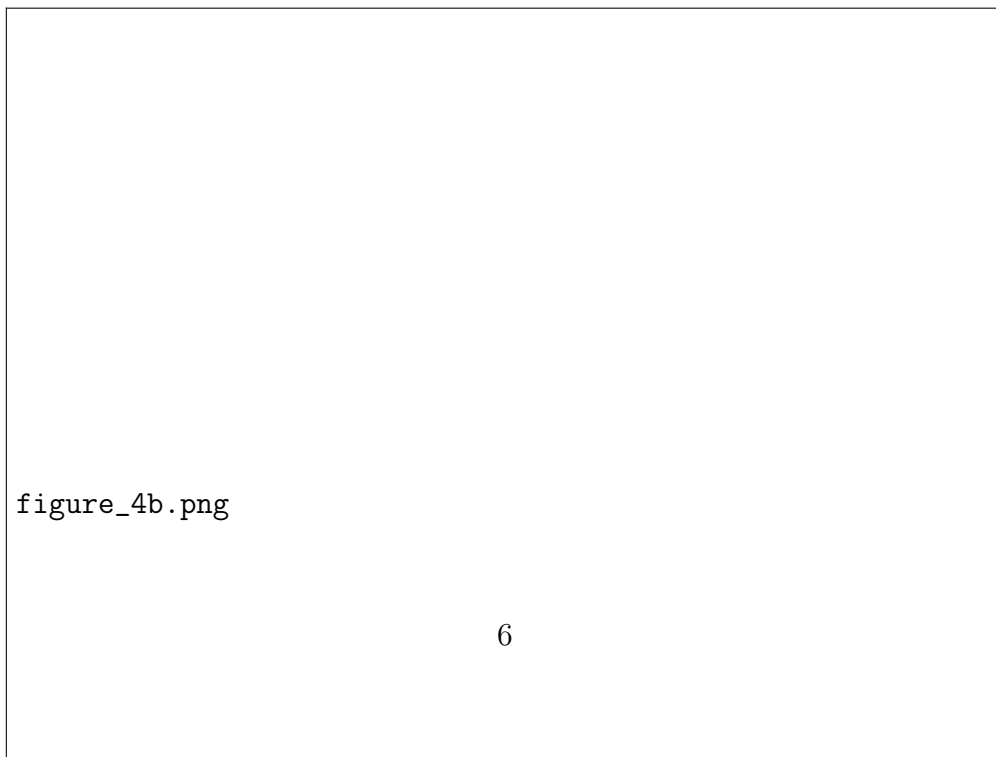
Figure 3: Pass-through of VAT exemptions to prices in Port Moresby by data source

Notes: Panel (a) plots normalized price indices for VAT-exempt (treatment) and non-exempt (control) items by data source, with prices scaled to equal one in May 2025. Panel (b) reports difference-in-differences estimates of the price effect of the VAT exemption reform, relative to May 2025. Estimates are based on Equation 1. The vertical line marks the introduction of the VAT exemptions on 1 June 2025, and the horizontal reference line denotes full pass-through of a 10 percent VAT removal. Vertical bars show 95 percent confidence intervals.

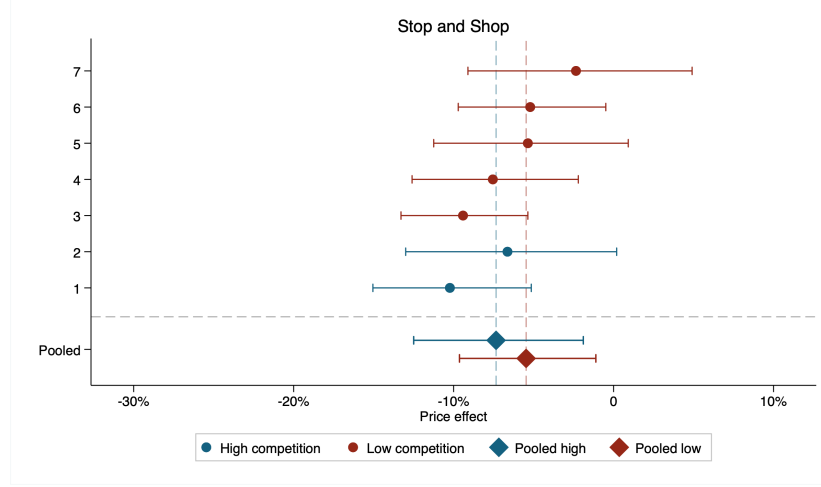
(a) Difference-in-Difference results by proximity



(b) Distribution of pass-through rates



(c) Differences in pass-through within the largest supermarket chain



(d) Differences in pass-through online vs in-store

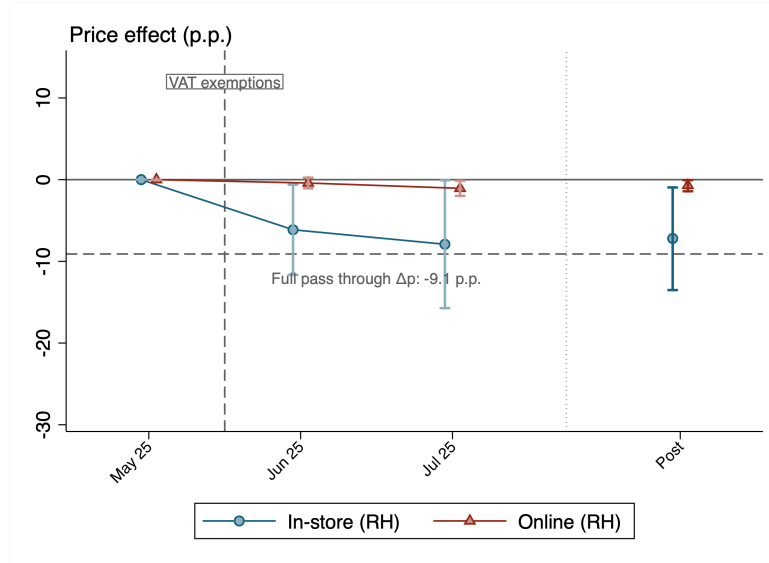


Figure 4: Heterogeneity of pass-through based on level of competition (continued)

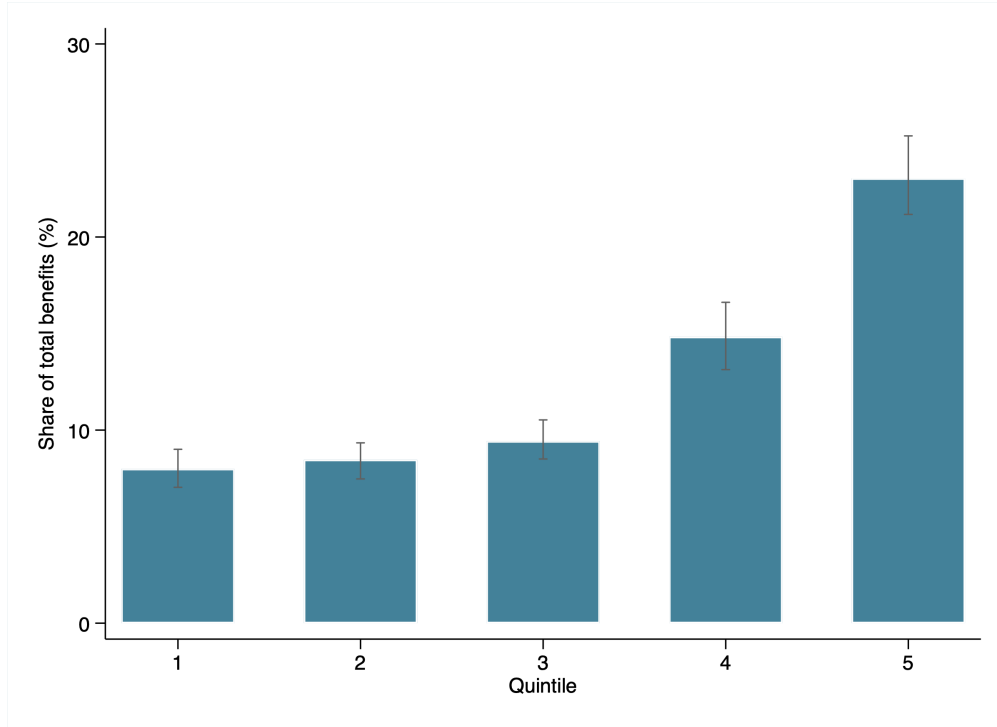
Notes: This figure examines heterogeneity in pass-through by local market competition using the supermarket census data. Panel (a) reports difference-in-differences estimates separately for stores in high-competition and low/mid-competition areas. Panel (b) shows the distribution of store-level pass-through rates three months after the reform. Panel (c) compares pass-through across stores within the largest supermarket chain, and Panel (d) compares online and in-store prices within the same chain. In Panels (a) and (d), the vertical line marks the reform date, and the dashed horizontal line indicates full pass-through of the VAT removal.

Table 2: VAT pass-through estimates (p.p.)

Pass-through, p.p.	(1)	(2)	(3)	(4)	(5)
Treatment	-3.1 (3.1)	-0.9 (2.6)	-0.0 (3.8)	-1.0 (2.5)	-1.0 (2.5)
Treatment $\times$ Formal	—	-4.9 (3.9)	-7.3* (4.3)	-1.3 (3.3)	-1.2 (3.3)
Treatment $\times$ Urban	—	—	—	—	1.2 (2.9)
Treatment $\times$ Formal $\times$ Urban	—	—	—	—	-6.7* (3.9)
Specification	Post	Post	Post	Post	Post
Coverage	All	All	Urban	Rural	All
N	129,921	129,921	43,725	86,196	129,921

Notes: Coefficients are in p.p. of price change for VAT-exempt vs. control items. Prices are normalized to item-specific May 2025 means. Post is the simple average of post-reform event-study coefficients from June 2025 to May 2026. Full pass-through corresponds to -9.1 p.p. Formal is based on the store type where the purchase occurred. Conventional standard errors are clustered at the product level and include the standard finite-sample correction. Significance stars are based on wild-cluster bootstrap p-values. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

(a) Share of foregone revenue accruing to households, by quintile



(b) Decomposition of consumer-side incidence

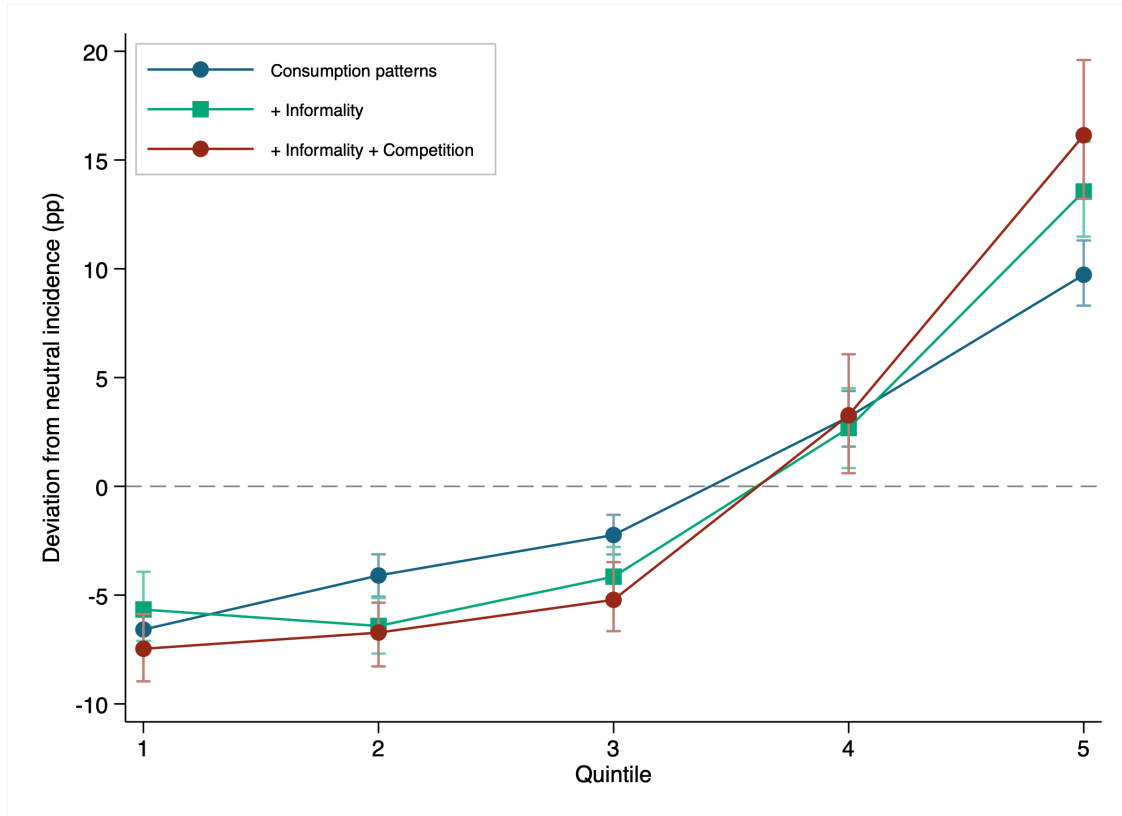


Figure 5: Incidence of VAT exemptions and decomposition of regressivity

Notes: Panel (a) reports the share of total foregone VAT revenue accruing to households in each expenditure quintile. The bars sum to 64 percent because the remaining share of foregone revenue is not passed through to households and is retained by stores and wholesalers. Panel (b) decomposes consumer-side incidence across quintiles. The blue series incorporates differences in consumption patterns only; the green

Table 3: Cost of delivering USD 12 million to the poorest quintile under alternative policies

Policy	Cost to deliver	Share of foregone revenue
Perfect targeting (Q1 only)	USD 12 million	8%
Universal transfer (no targeting)	USD 60 million	40%
Universal transfer (50% leakage)	USD 120 million	80%
Child grant and nutrition program	USD 150 million	100%
VAT exemption (actual)	USD 150 million	100%

Notes: This table compares the fiscal cost of alternative policies that would deliver the same monetary benefit to the poorest quintile as the VAT exemption reform. The benchmark assumes that the actual VAT exemption delivers approximately USD 12 million to the poorest quintile, or about 8 cents of each dollar of foregone revenue. The alternative rows report stylized transfer schemes under different targeting assumptions and implied leakage.

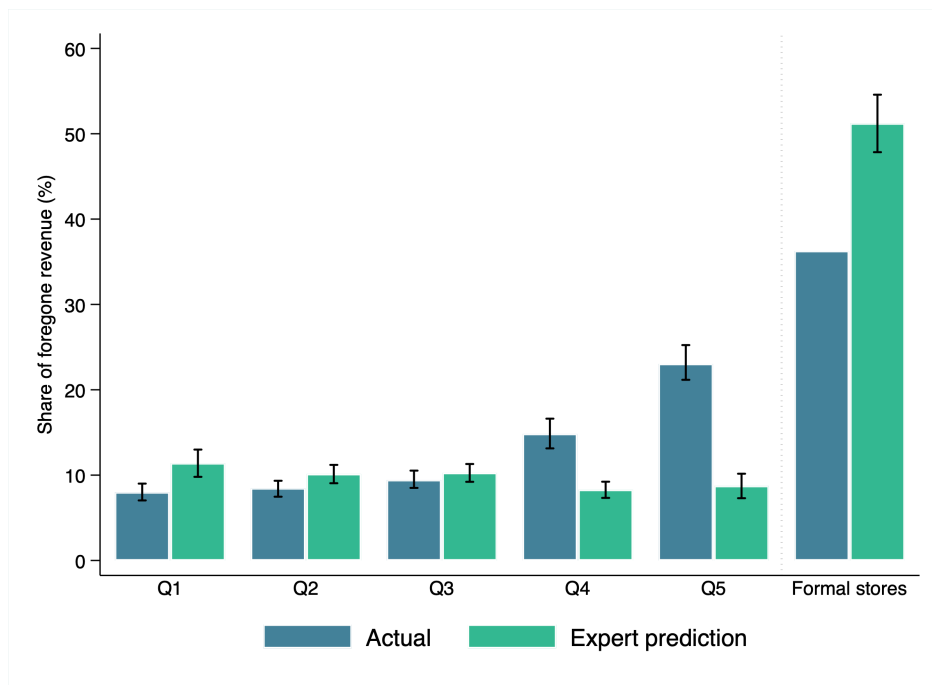


Figure 6: Actual incidence vs. expert predictions of revenue lost across quintiles and stores

Notes: This figure compares the realized incidence of the VAT exemption reform with experts' ex ante predictions. Bars show the share of foregone revenue accruing to formal stores and to each household expenditure quintile. Realized incidence is computed from the phone survey using observed consumption patterns and estimated pass-through, while predicted incidence is based on the expert survey. The figure highlights the gap between expected and actual distributional outcomes.

Appendix A – Additional Figures and Tables

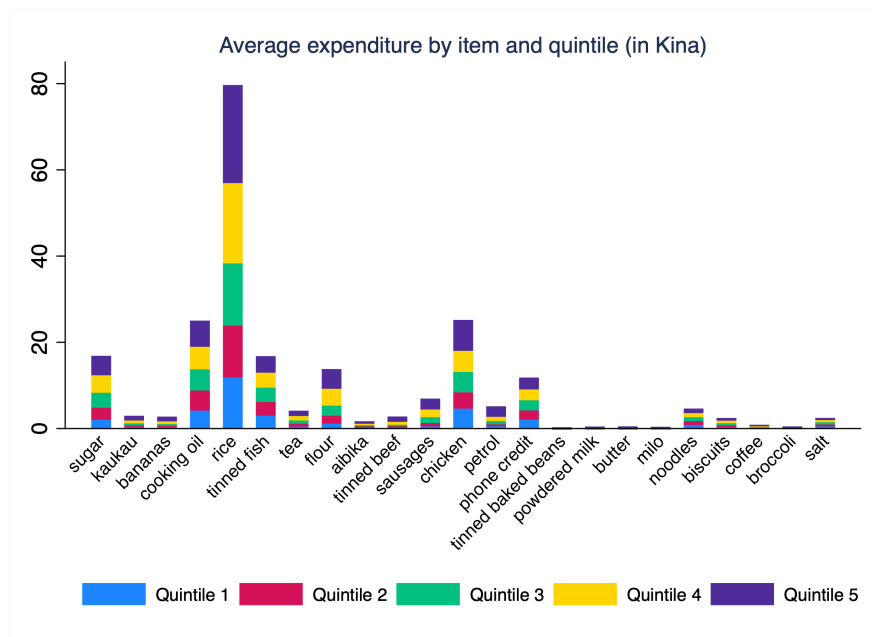


Figure A1: Average expenditure by item per quintile

Note: This figure shows the average expenditure by item per quintile in Kina.

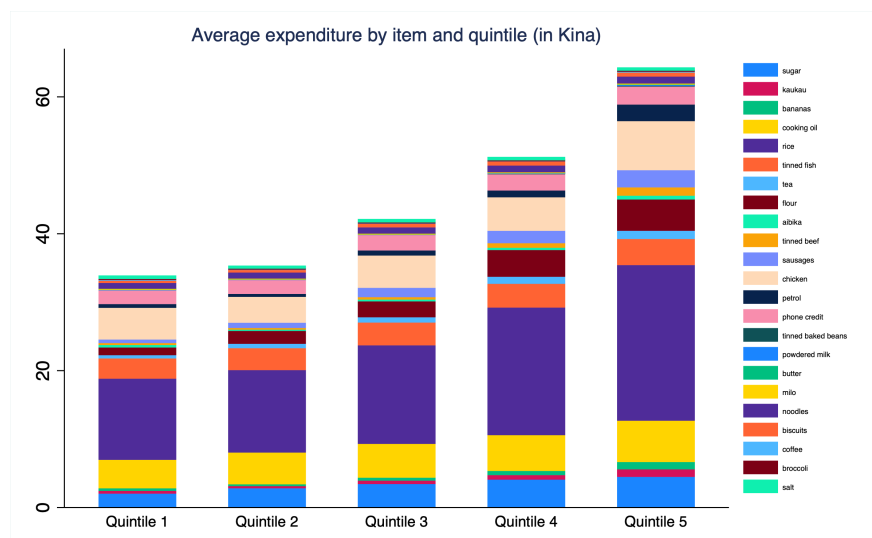


Figure A2: Average expenditure by quintile per item

Note: This figure shows the average expenditure by quintile per item in Kina.

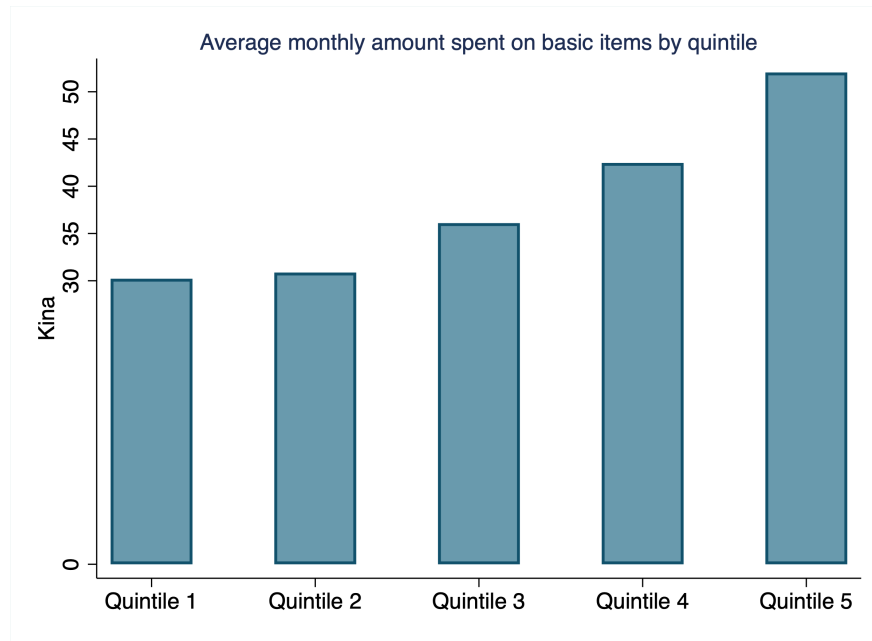


Figure A3: Average expenditure by quintile on basic items

Note: This figure shows the average expenditure by quintile on basic items in Kina.

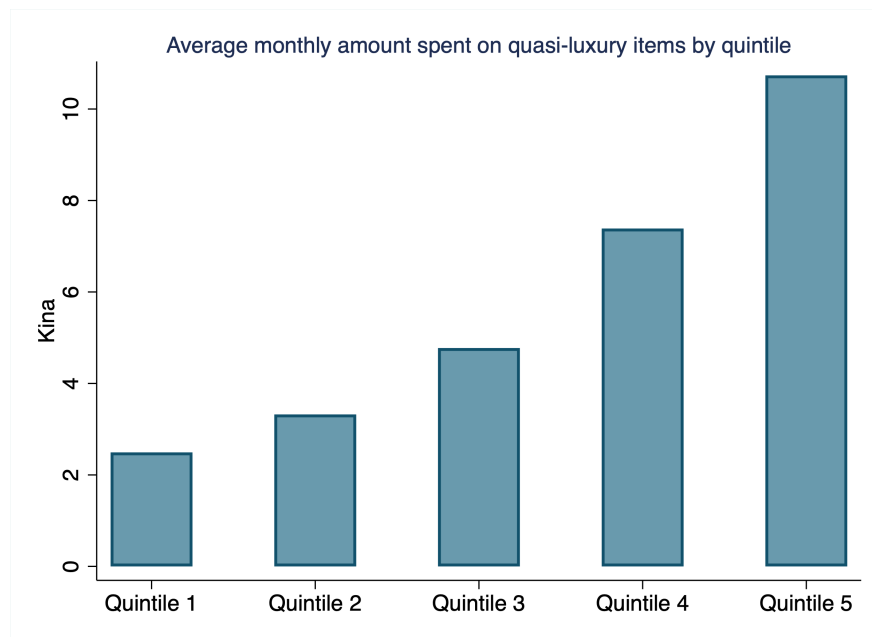


Figure A4: Average expenditure by quintile on quasi-luxury items

Note: This figure shows the average expenditure by quintile on quasi-luxury items in Kina.

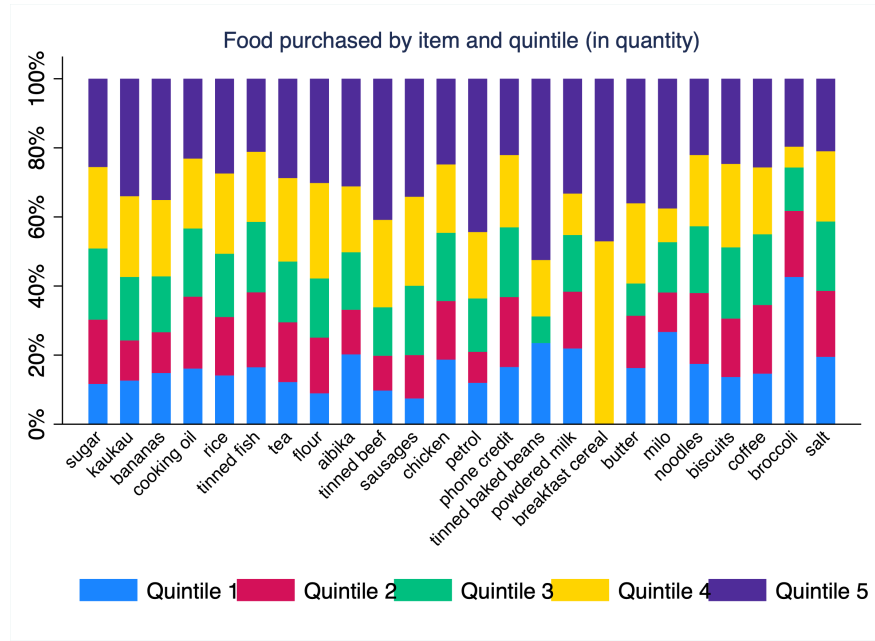


Figure A5: Quantity purchased by item per quintile

Note: This figure shows the quantity purchased by item per quintile.

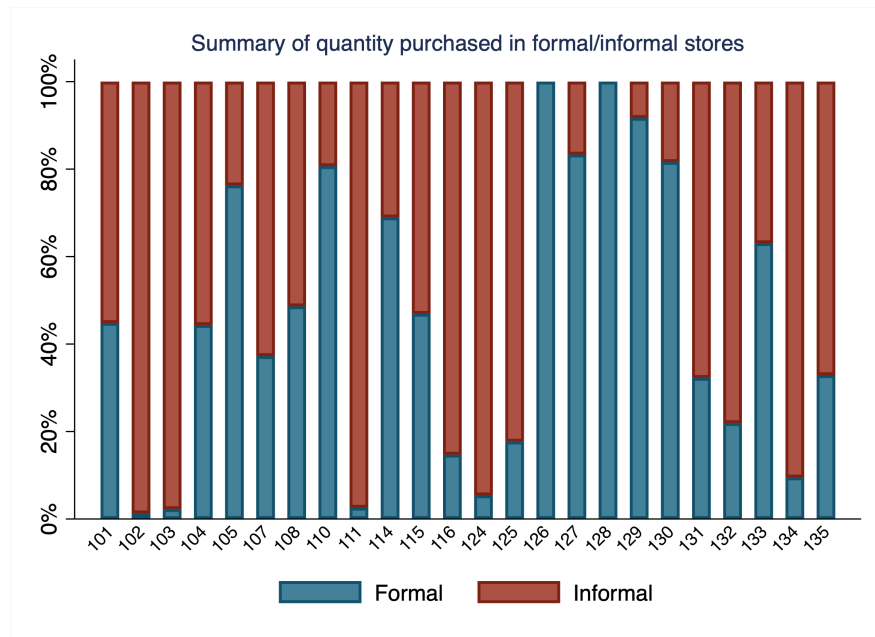


Figure A6: Quantity purchased by item by store type

Note: This figure shows the quantity purchased by item by store type.

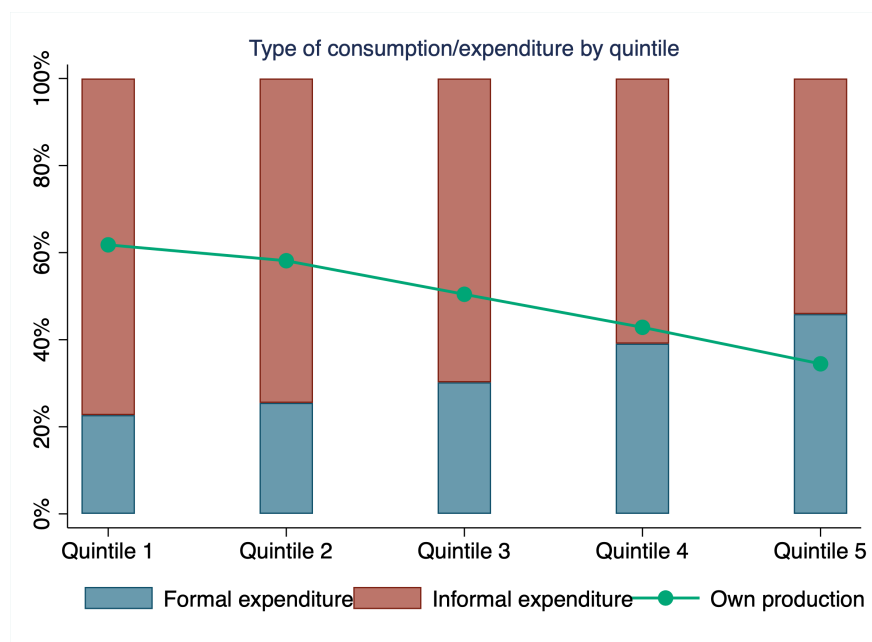


Figure A7: Type of consumption and expenditure by quintile

Note: This figure shows the type of consumption and expenditure by quintile based on 2024 data from the phone survey.

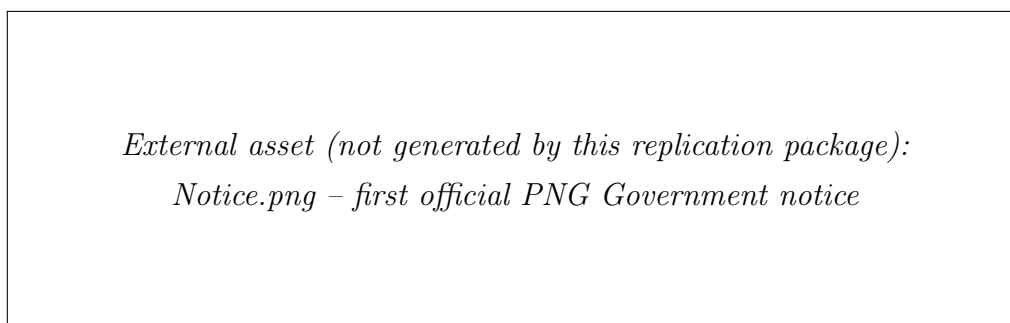


Figure A8: First official notice from the PNG Government about the VAT exemption reform

Note: This figure shows the first official notice from the PNG Government about the VAT exemption reform.

Table A1: Timeline of the VAT Exemptions Reform and Study Data Collection

Date	Event
Oct 2023 to May 2026	Nationally representative, monthly phone survey panel (about 1,000 households/month) measuring quantities, expenditures, and store type (formal/informal) for 20 items.
Dec 2023 to May 2026	Administrative data collected monthly by (i) Central Bank of PNG research team and (ii) National Statistics Office CPI team; providing detailed price data for over 100 items from a subset of formal supermarkets in Port Moresby.
29 Nov 2024	PNG government first publicly stated the <i>possibility</i> of removing VAT from some basic items in mid-2025; details were limited and credibility unclear.
19 May 2025	Policymakers announced that VAT exemptions for 10 food items (and 3 non-food items) would take effect on June 1, 2025 (i.e., 12 days later).
May-Aug 2025	Weekly web-scraped all online prices from the two chains offering online shopping in Port Moresby (including the use of “time machine” software to check prices before reform announced).
May-Nov 2025	Monthly in-store supermarket census collected prices for 20 food items (10 exempt and 10 non-exempt) in all 83 formal supermarkets in Port Moresby.
1 June 2025	Reform implemented: VAT on 10 food items set to zero-rated (functionally close to an exemption in practice). Exemptions scheduled to remain in place for at least 13 months.
July-Sep 2025	Expert prediction survey hosted on SSPP to benchmark beliefs about pass-through and incidence against realized outcomes.
8 Aug 2025	Policymakers issued revised guidance clarifying the details of the reform and outlining plans to monitor its impact on prices.
25 Nov 2025	PNG government announced an extension of the VAT exemption policy to the end of 2026 (i.e., 19 months in total).
Dec 2025	Phone survey question about how familiar households are with the details of the reform six months after implementation.

Notes: The table lists both policy milestones and the data-collection windows used to estimate pass-through and incidence.

*External asset (not generated by this replication package):
Figure_1_Page_1 / _2 / _3 – second official PNG Government notice (3
pages)*

Figure A9: Second official notice from the PNG Government about the VAT exemption reform

Note: This figure shows the second official notice from the PNG Government about the VAT exemption reform.

	Survey			Census
	All months	1-month	Monthly average	
BOUGAINVILLE	4.81	5.76	4.62	3.51
CENTRAL	3.00	2.49	2.73	3.01
CHIMBU	5.52	7.20	5.73	5.54
EAST NEW BRITAIN	4.93	5.10	4.51	4.25
EAST SEPIK	5.84	6.02	5.91	6.36
EASTERN HIGHLANDS	8.68	7.72	7.72	9.97
ENGA	4.65	5.63	5.50	5.56
GULF	2.01	1.96	2.45	1.88
HELA	3.23	5.10	4.25	4.75
JIWAKA	4.62	4.97	5.33	4.74
MADANG	6.07	5.24	6.24	6.27
MANUS	2.49	2.62	2.52	0.75
MILNE BAY	2.37	1.83	2.10	4.02
MOROBE	8.28	6.54	7.62	9.46
NATIONAL CAPITAL DISTRICT	5.68	5.89	4.15	3.46
NEW IRELAND	3.43	2.62	3.45	2.16
NORTHERN (ORO)	2.09	2.09	2.55	2.48
SOUTHERN HIGHLANDS	4.93	6.54	6.34	6.40
WEST NEW BRITAIN	4.50	2.88	3.69	3.69
WEST SEPIK (SANDAUN)	4.54	3.66	4.07	3.27
WESTERN	1.78	1.05	1.54	2.28
WESTERN HIGHLANDS	6.55	6.81	6.98	6.19

Table A2: Distribution of survey respondents relative to population

Note: This table shows the distribution of survey respondents relative to the population.

Table A3: Familiarity with National Government's Efforts to Reduce Tax

Response	Freq.	Percent	Cum.
Not at all	510	38.06	38.06
A little	536	40.00	78.06
Somewhat	248	18.51	96.57
Very Familiar	46	3.43	100.00
Total	1,340	100.00	

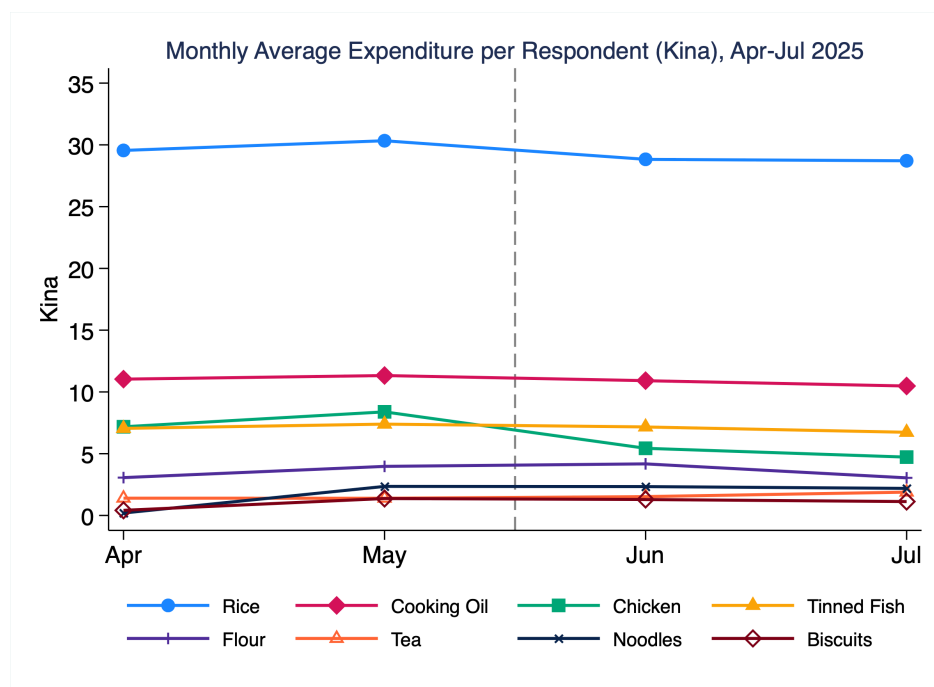


Figure A10: Average monthly expenditure on exempt items

Note: This figure shows the average monthly expenditure on exempt items.

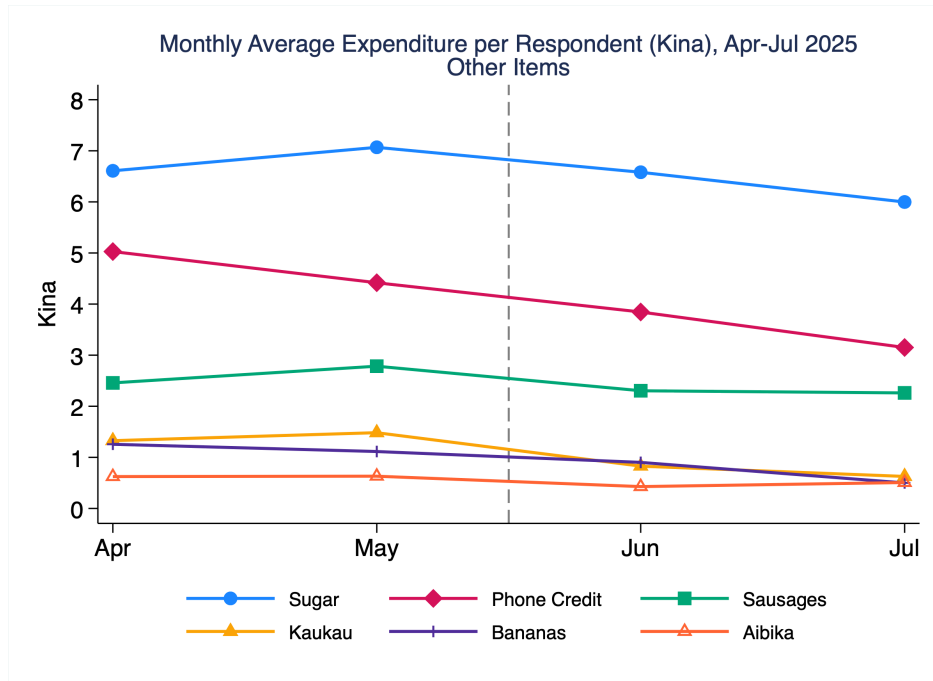


Figure A11: Average monthly expenditure on non-exempt items

Note: This figure shows the average monthly expenditure on non-exempt items.

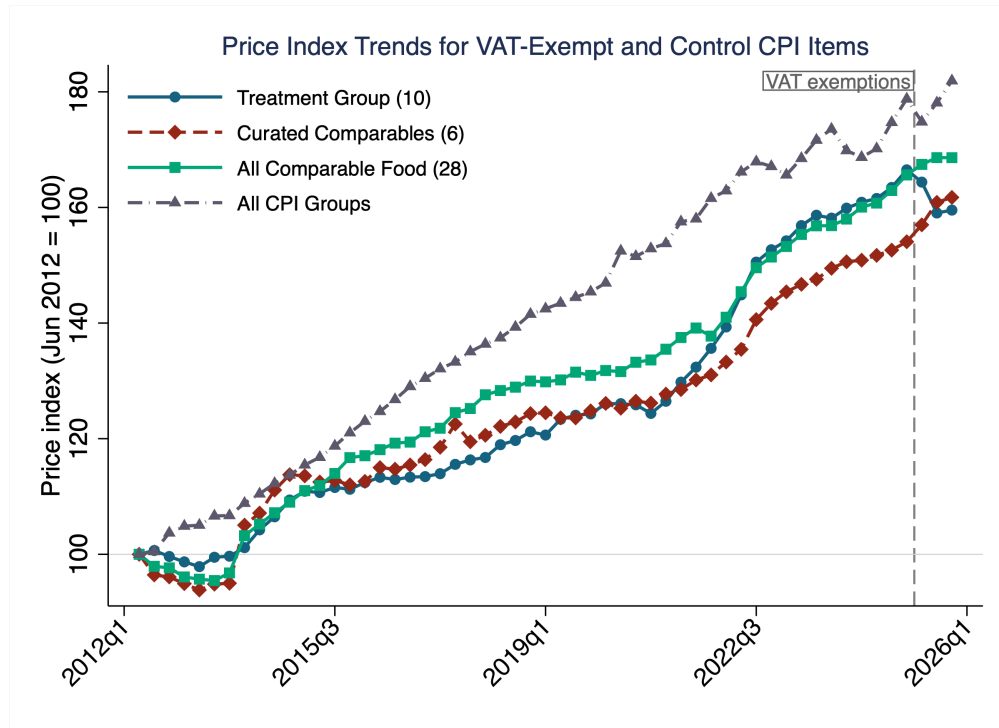


Figure A12: Raw trends in prices over the last 14 years according to NSO administrative data

Note: This figure shows the raw trends in prices over the last 14 years according to NSO administrative data.

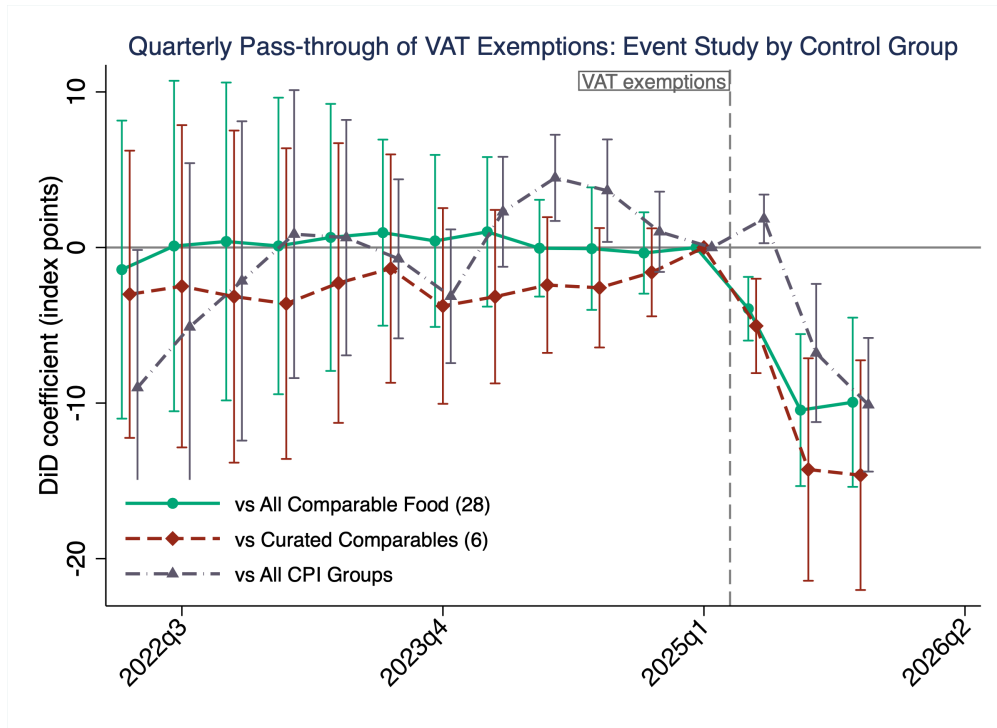


Figure A13: Differences-in-Differences results with different comparison groups over the last four years according to NSO administrative data

Note: This figure shows the Differences-in-Differences results with different comparison groups over the last four years according to NSO administrative data.

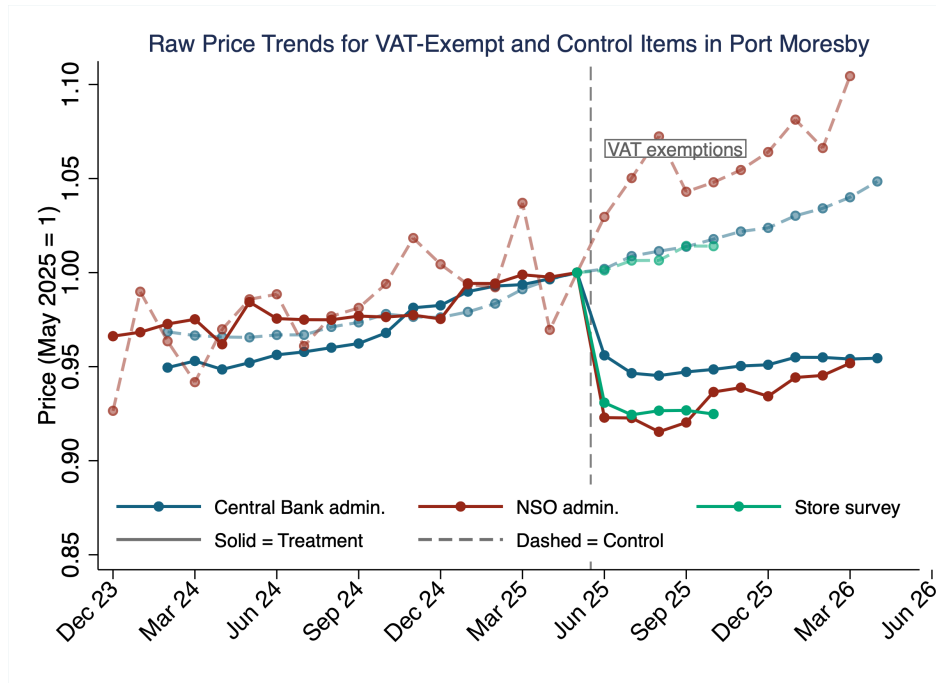


Figure A14: Raw trends of prices in Port Moresby by data source

Note: This figure shows the raw trends results by data source.

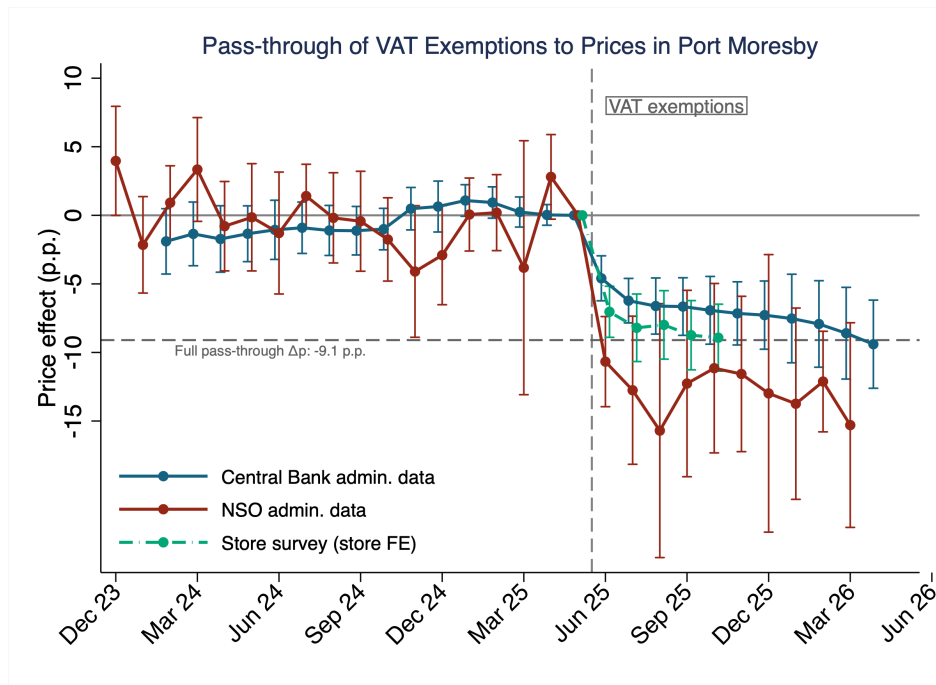


Figure A15: Pass-through of VAT exemptions to prices in Port Moresby by data source

Note: This figure shows the difference-in-differences results by data source.

*External asset (not generated by this replication package):
shop.jpg – Stop and Shop advertisement*

Figure A16: Advertisement from Stop and Shop

Note: This figure presents an advertisement from the largest supermarket chain, Stop and Shop.

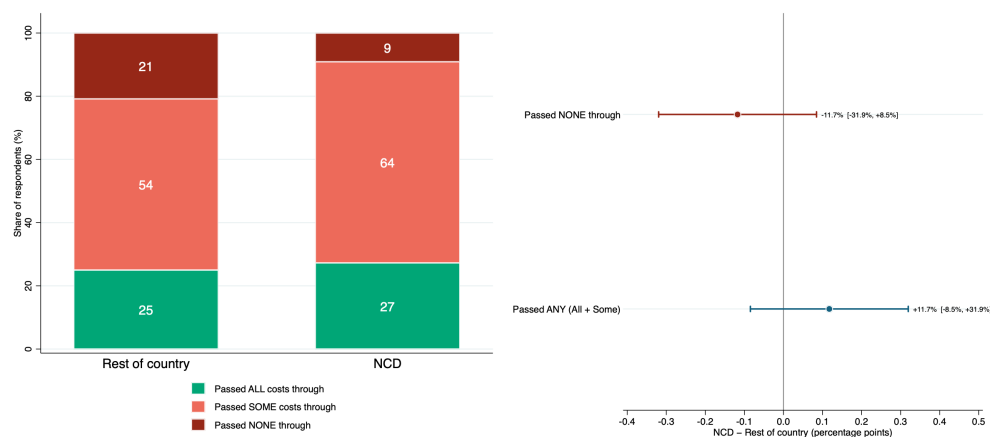


Figure A17: Results of pass-through question in business sentiment survey

Note: This figure presents the results of the pass-through question in the business sentiment survey administered by the central bank.

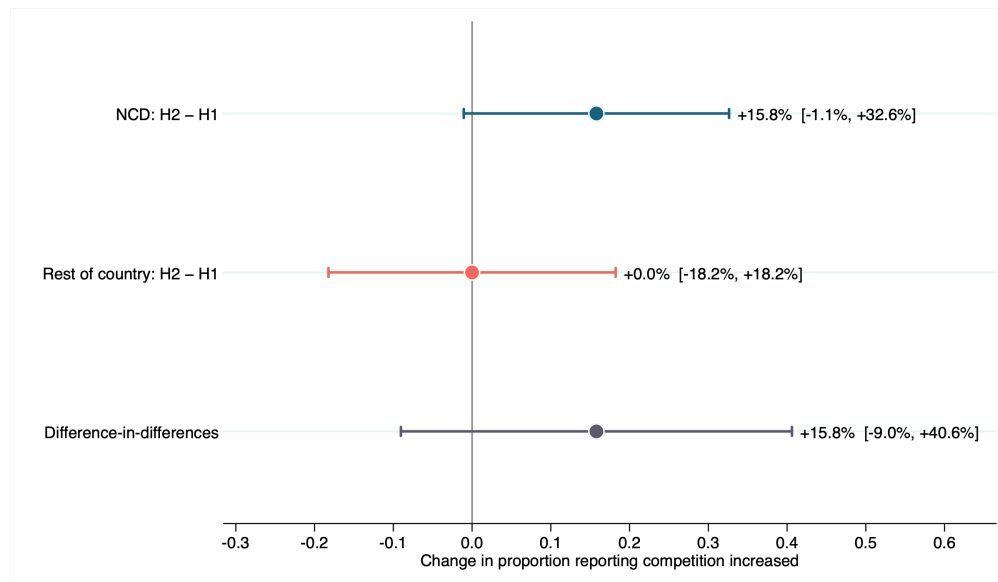


Figure A18: Results of competition question in business sentiment survey

Note: This figure presents the results of the competition question in the business sentiment survey administered by the central bank.

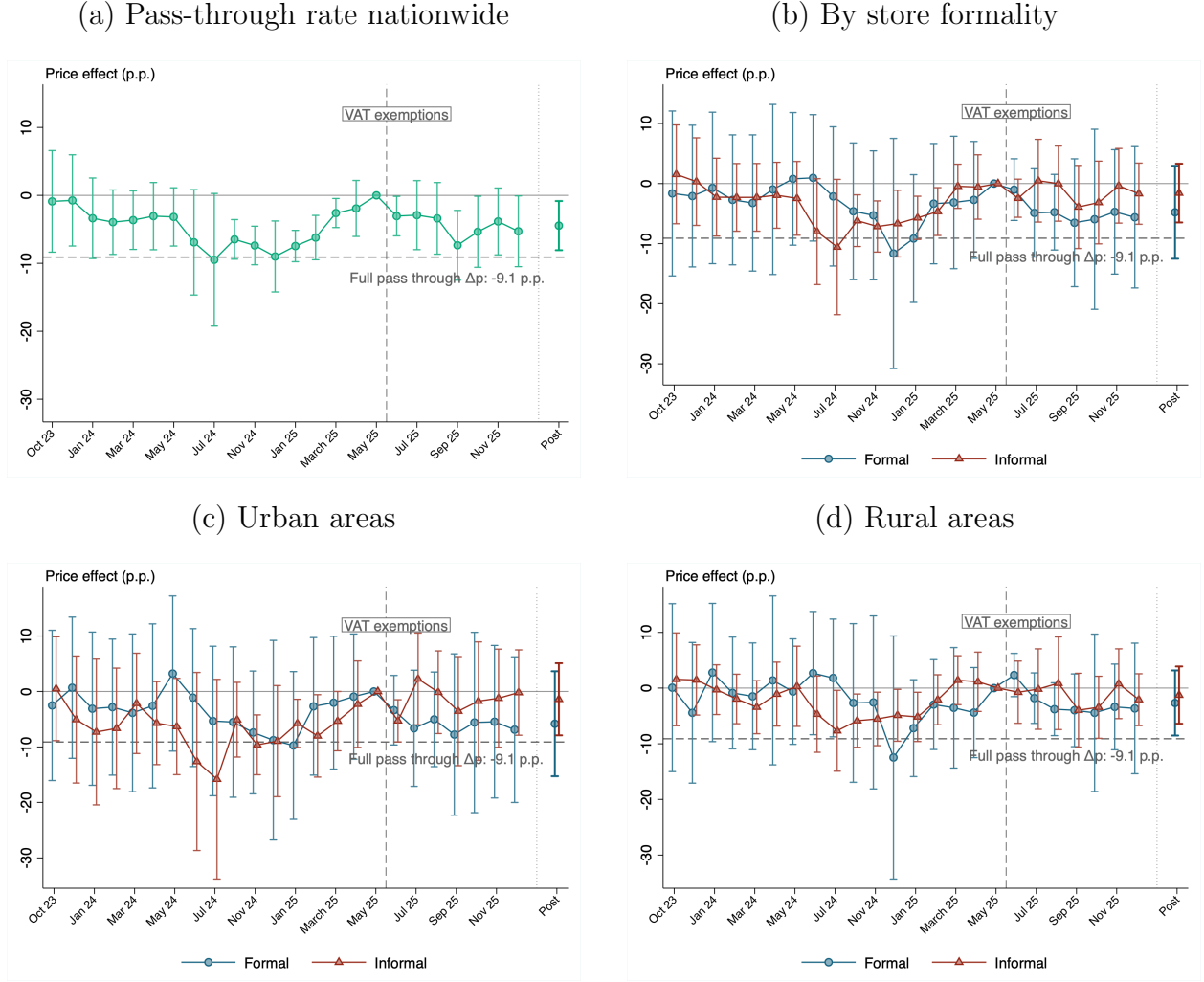


Figure A19: Pass-through nationwide based on the phone survey data

Notes: This figure reports difference-in-differences estimates using the nationally representative phone survey. Panel (a) shows the average pass-through nationwide. Panel (b) disaggregates by store formality. Panels (c) and (d) further split the results by urban and rural location. The vertical line marks the June 2025 reform, and the dashed horizontal line indicates full pass-through of the VAT removal.

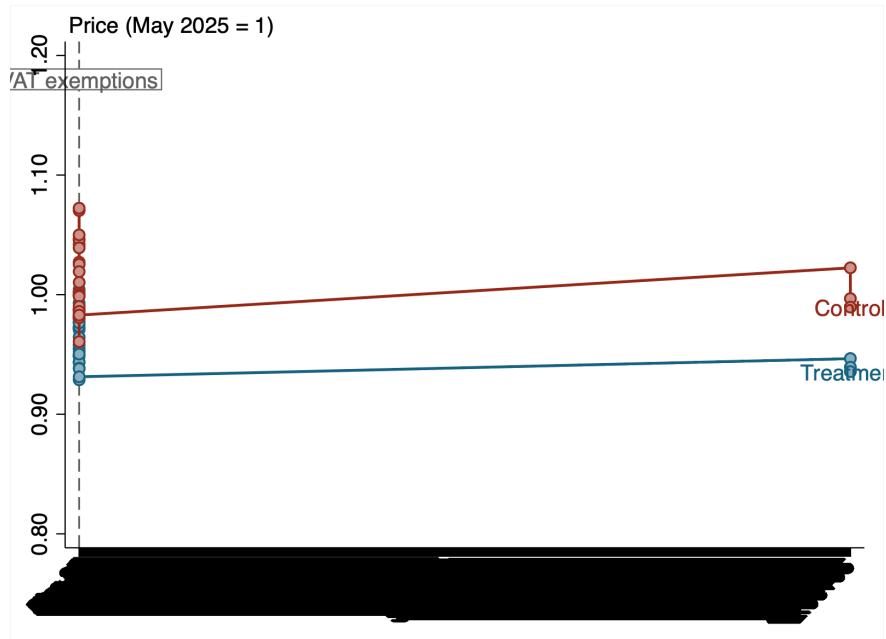


Figure A20: Raw trends in prices in treatment and control groups in the phone survey data

Note: This figure shows the raw trends in prices in treatment and control groups in the phone survey data.

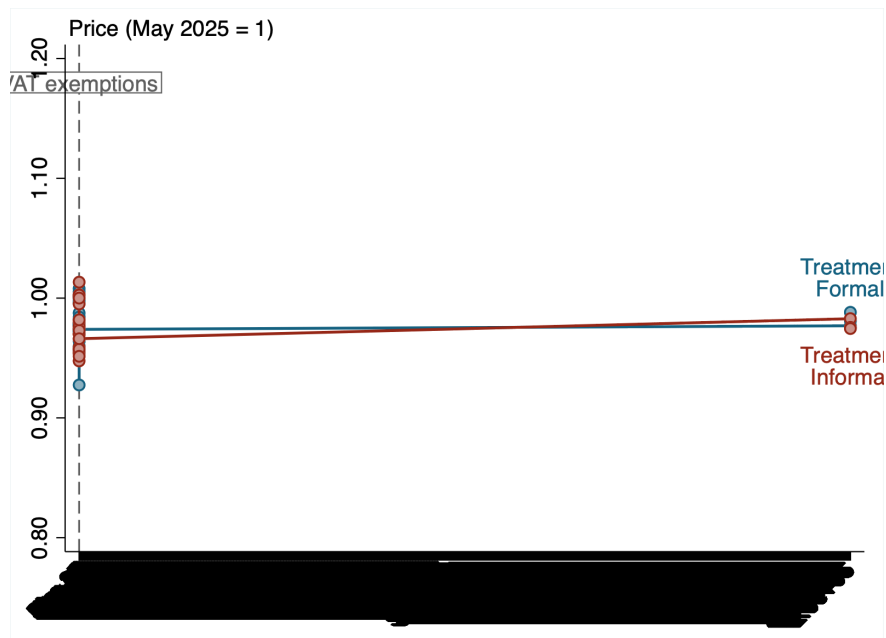


Figure A21: Raw trends in prices in treatment group between formal and informal stores in the phone survey data

Note: This figure shows the raw trends in prices in treatment group between formal and informal stores in the phone survey data.

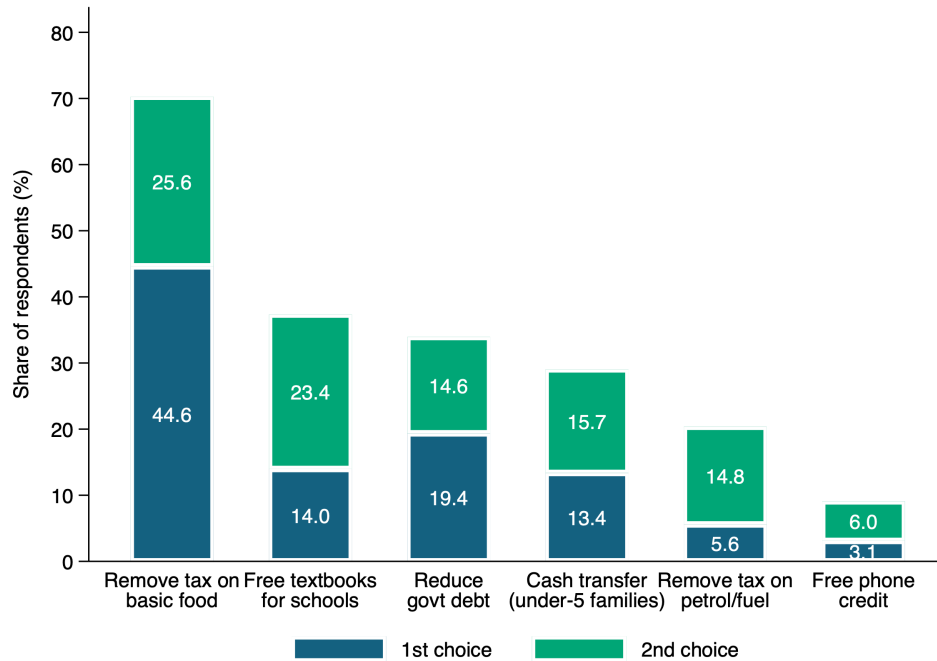


Figure A22: Preferences for alternative policies

Note: This figure shows the results of a question about preferences for alternative policies asked in the April 2026 round of the phone survey. The exact wording was as follows (and respondents selected their first and second preference): “Imagine the government was considering introducing a program to support poor and middle-class families. In your view, which of the following would be the best way to support poor and middle-class families? provide a cash transfer for all families with children under the age of five; remove taxes from basic food items sold in supermarkets; remove taxes from petrol and other fuels; provide free textbooks for schools; provide a small amount of free phone credit for all phone users; reduce the level of government debt.”

Table A4: Treatment effects on support for removing the tax on food items, by prior policy preference

Subgroup	N	Unweighted	Weighted
		(robust SE)	(robust SE)
Overall (ATE)	995	−0.119* (0.063)	−0.106 (0.091)
<i>Preferred way to support households (April 2026):</i>			
Prefers removing food tax	340	+0.049 (0.113)	+0.186 (0.151)
Prefers another policy	412	−0.230** (0.090)	−0.285** (0.119)
<i>Another policy, by type:</i>			
Cash transfer (under-5s)	86	−0.796*** (0.200)	−0.914*** (0.288)
Reduce government debt	152	−0.036 (0.151)	−0.207 (0.213)
Free textbooks	109	−0.285* (0.173)	−0.199 (0.197)
Remove tax on fuel	46	+0.200 (0.262)	−0.019 (0.242)
Free phone credit	19	—	—
<i>Interaction tests:</i>			
Prefers food tax × Treat		+0.286** (0.143)	+0.467** (0.192)
Prefers cash × Treat		−0.775*** (0.212)	−0.927*** (0.311)
Joint test, all groups (<i>F</i>)		<i>p</i> = 0.007	<i>p</i> = 0.049

Notes: The outcome is support for removing the tax on selected food items, on a 1–5 scale (5 = strongly support; control-group mean 3.76). Cells report subgroup average treatment effects, conditioning on the orthogonal question-version indicator, with heteroskedasticity-robust (HC1) standard errors in parentheses. The unweighted estimates (preferred specification) are shown alongside weighted estimates that apply the household survey weights as a robustness check. Treatment was randomly assigned in the May 2026 round ($N = 995$); respondents’ preferred way to support households is taken from the same households’ April 2026 responses, which restricts the heterogeneity analysis to the 752 households observed in both rounds. The interaction rows report the treatment × moderator coefficient (food tax versus all other preferences; cash transfer versus all others); the final row reports the *p*-value of a joint *F*-test across the six preference groups. The smallest group (free phone credit) is too small to estimate separately but is retained in the joint test. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Finished (%)	73.84
Median duration (minutes)	8.97
Researcher (%)	62.44
Economics (%)	92.44
Research experience in:	
Public Finance (%)	52.54
Tax Incidence (%)	31.07
Development (%)	58.76
Industrial Organization (%)	16.38
Low-middle income countries (%)	62.15
Number of responses	237
Number of complete responses	175

Table A5: Background information about expert prediction survey

Note: This table shows background information about the expert prediction survey.

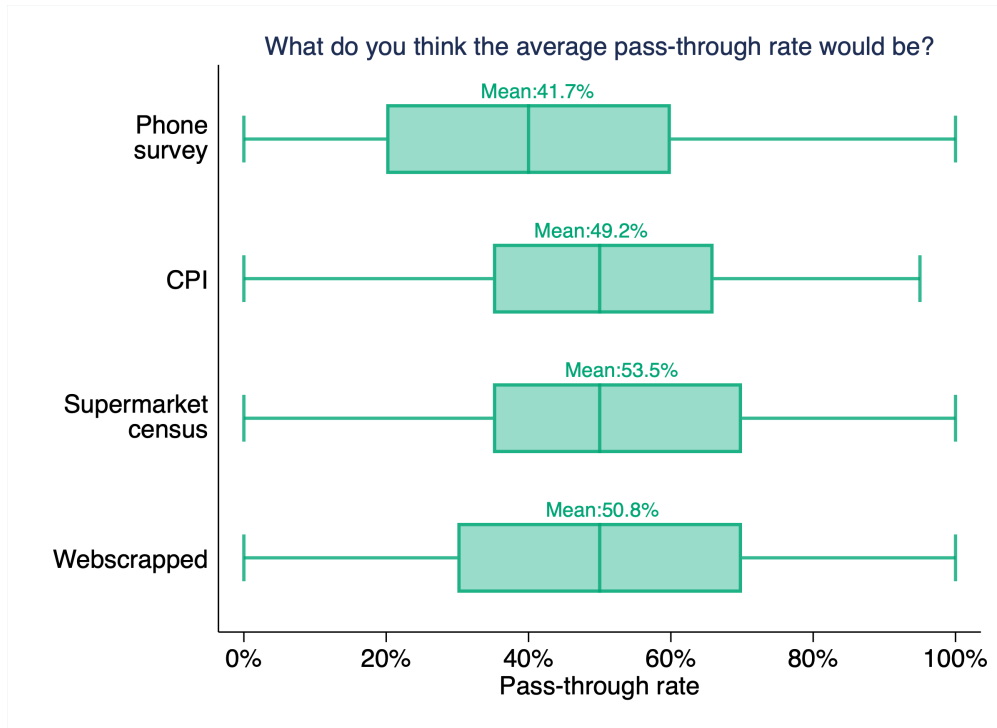


Figure A23: Experts' predictions of pass-through by data source

Note: This figure shows experts' predictions of pass-through by data source.

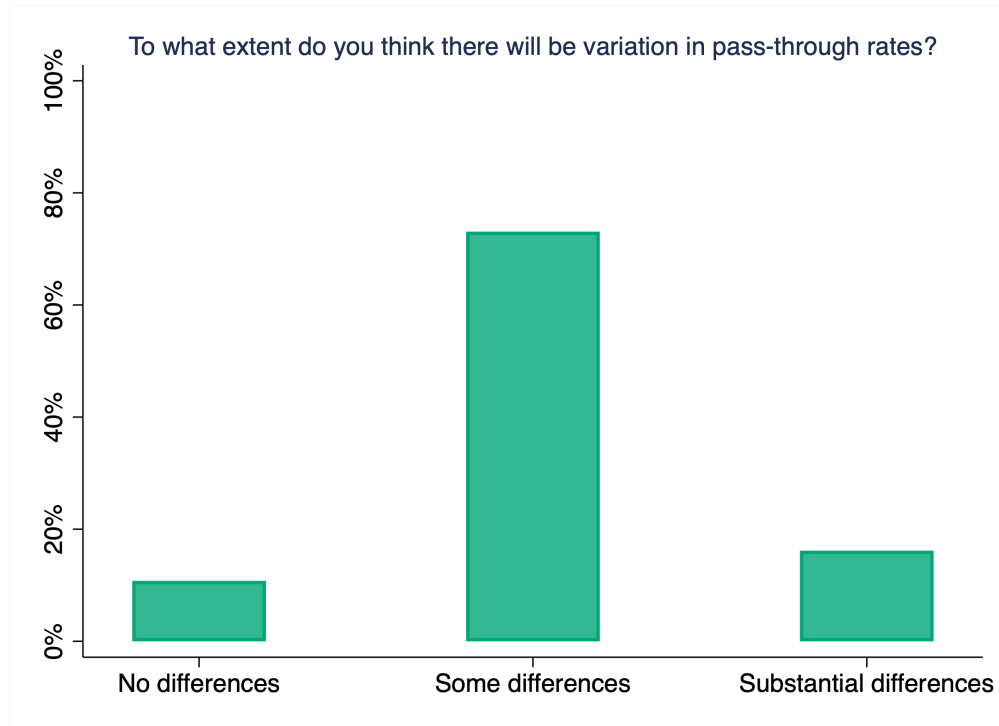


Figure A24: Experts' predictions of variation in pass-through

Note: This figure shows experts' predictions of variation in pass-through.

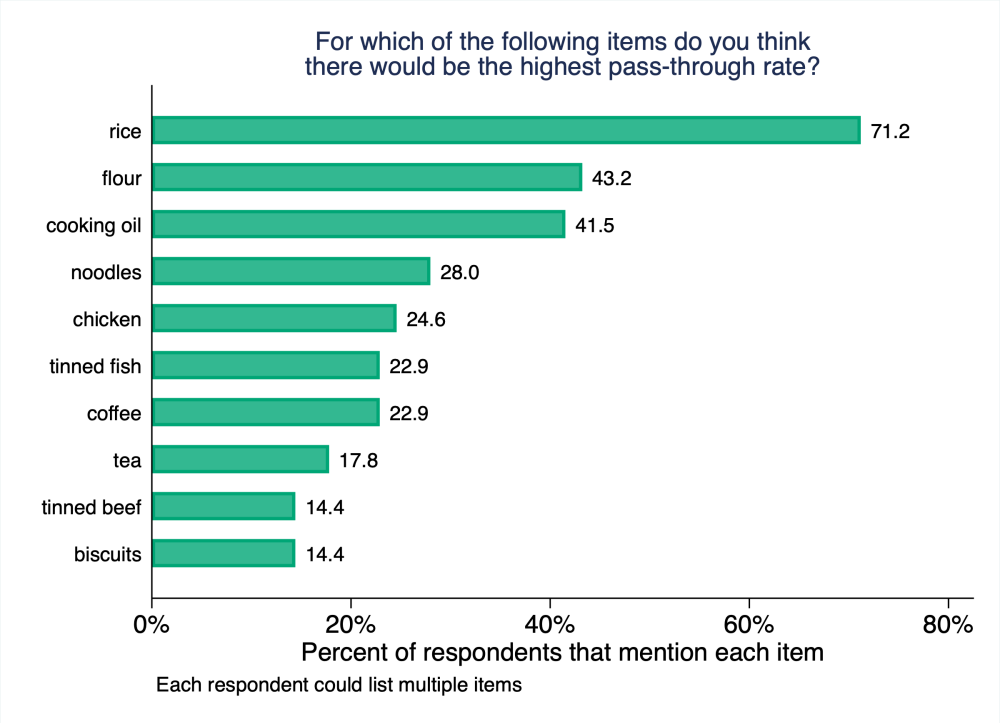


Figure A25: Experts' predictions of items with high pass-through

Note: This figure shows experts' predictions of items with high pass-through.

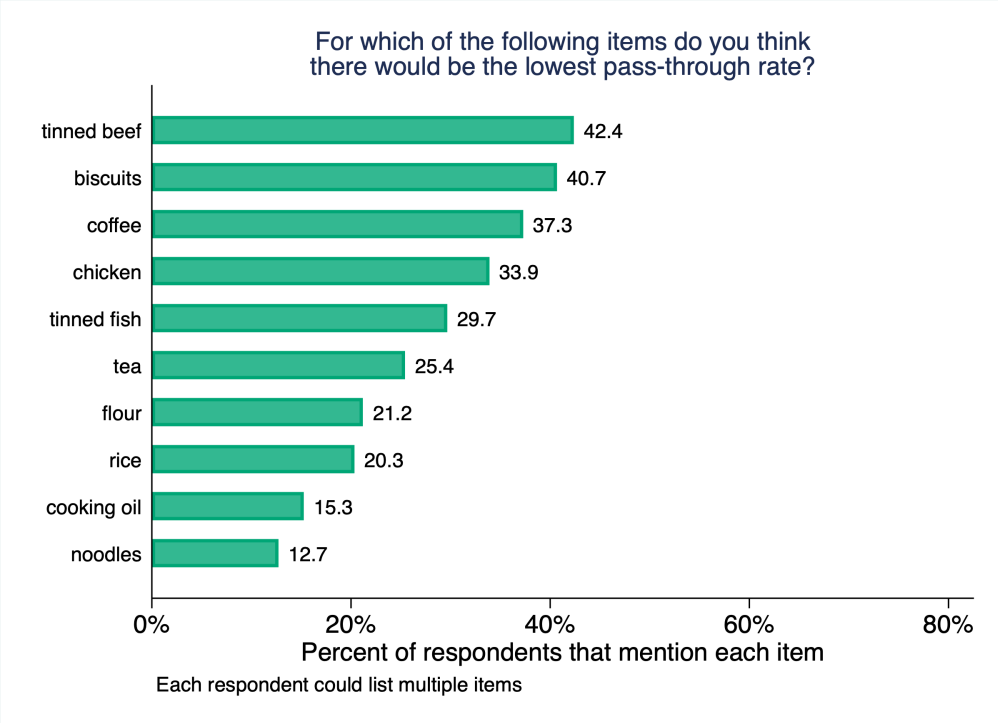


Figure A26: Experts' predictions of items with low pass-through

Note: This figure shows experts' predictions of items with low pass-through.

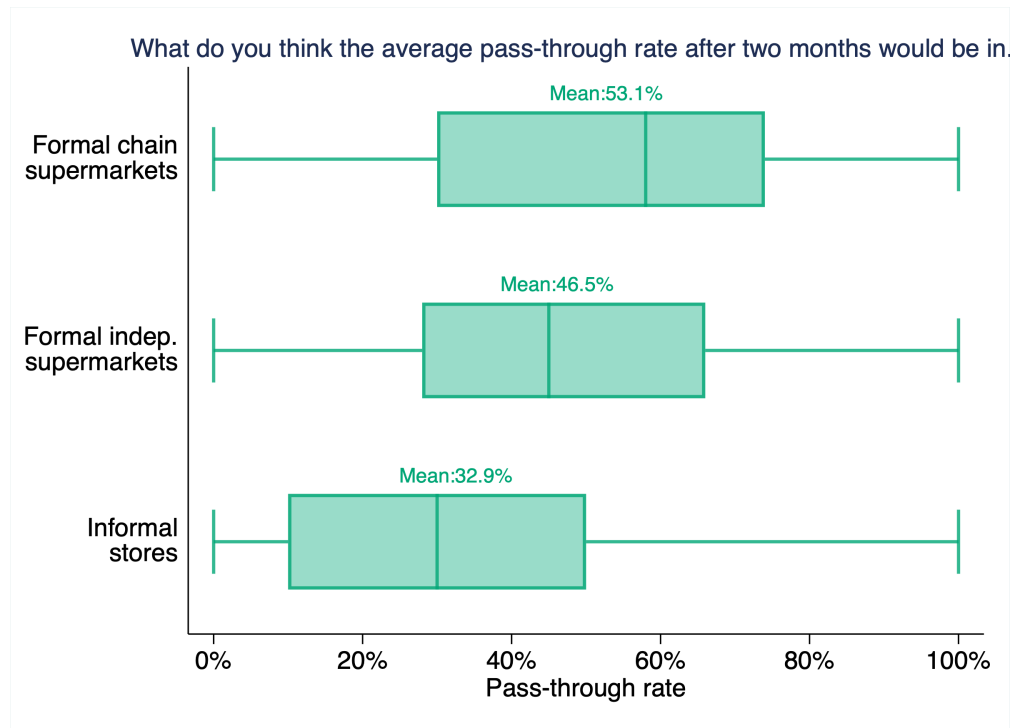


Figure A27: Experts' predictions of pass-through by store type

Note: This figure shows experts' predictions of pass-through by store type.

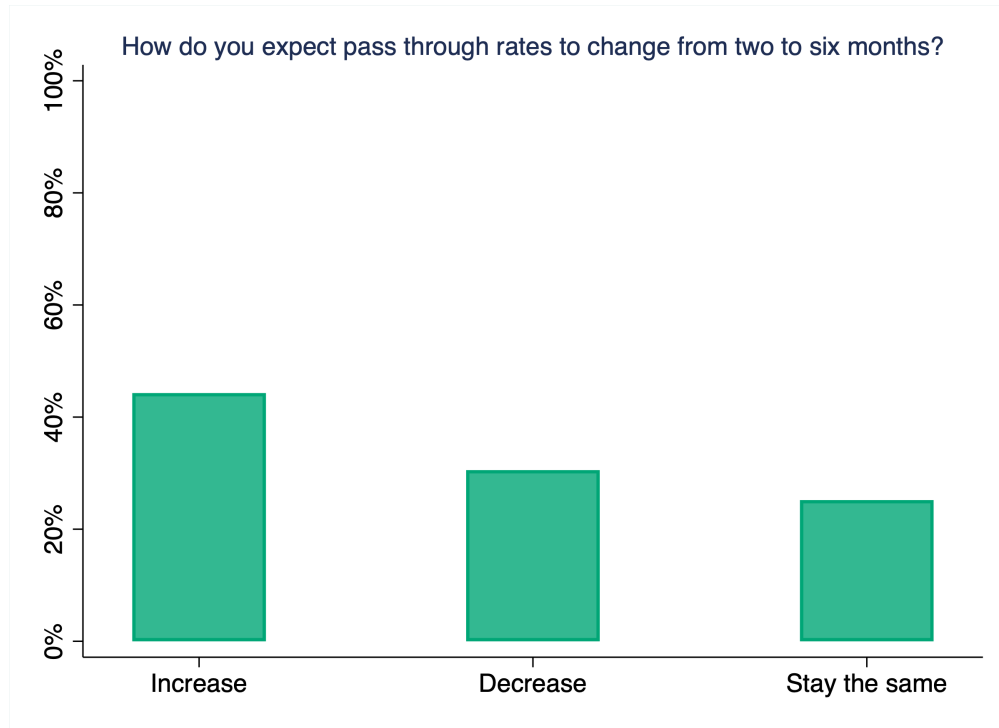


Figure A28: Experts' predictions of changes in pass-through over time

Note: This figure shows experts' predictions of changes in pass-through over time.

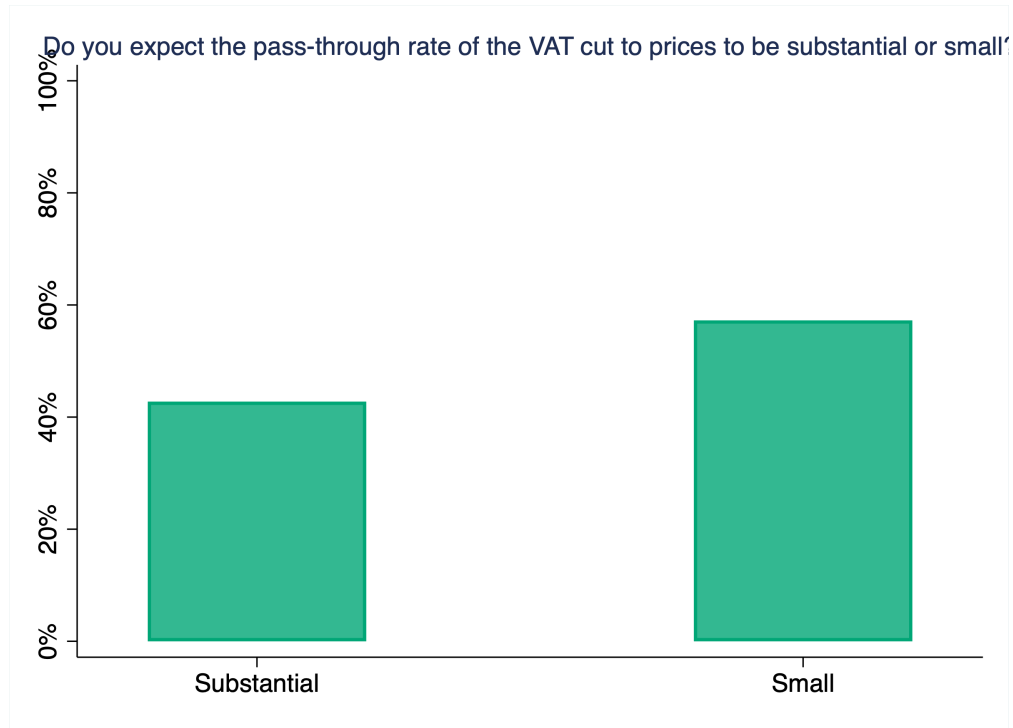


Figure A29: Experts' predictions of size of pass-through

Note: This figure shows experts' predictions of size of pass-through.

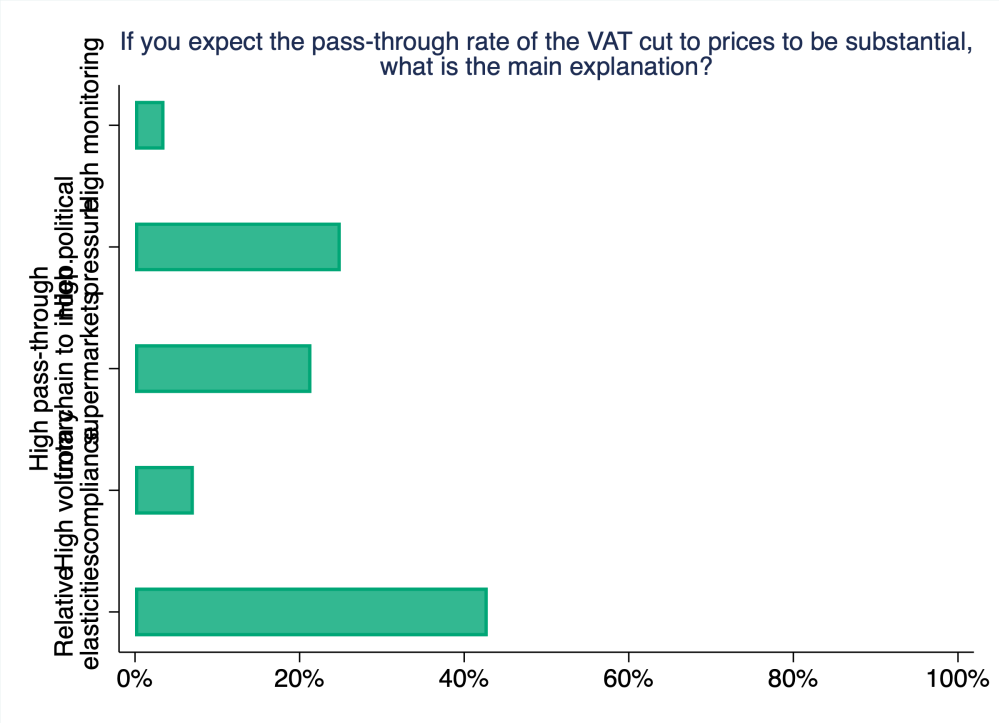


Figure A30: Experts' reasons for substantial pass-through

Note: This figure shows experts' reasons for substantial pass-through.

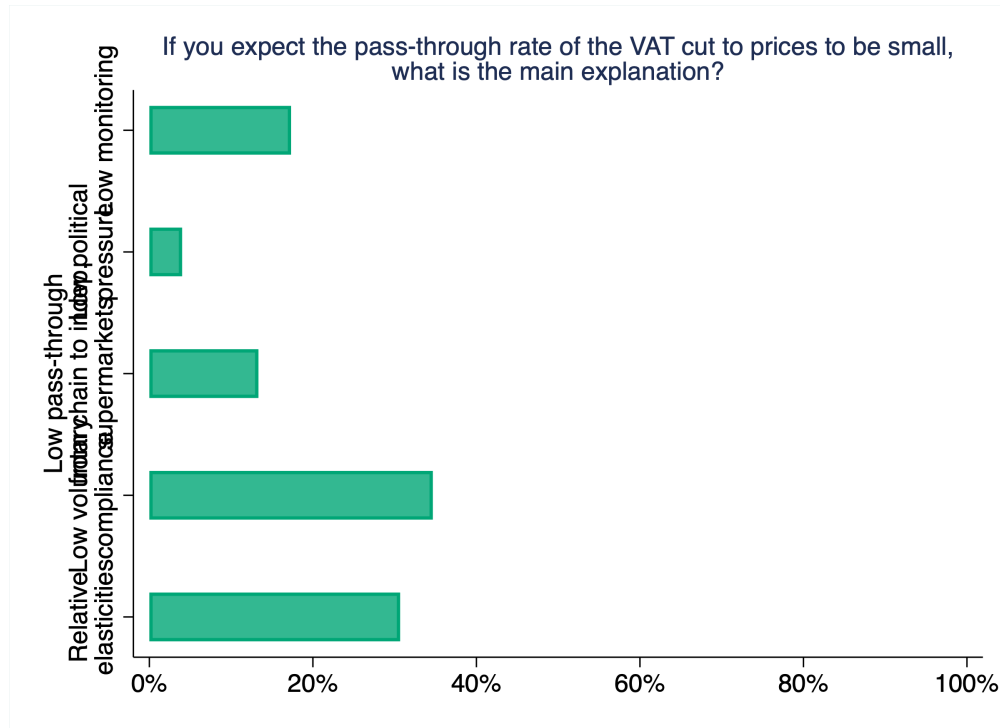


Figure A31: Experts' reasons for small pass-through

Note: This figure shows experts' reasons for small pass-through.

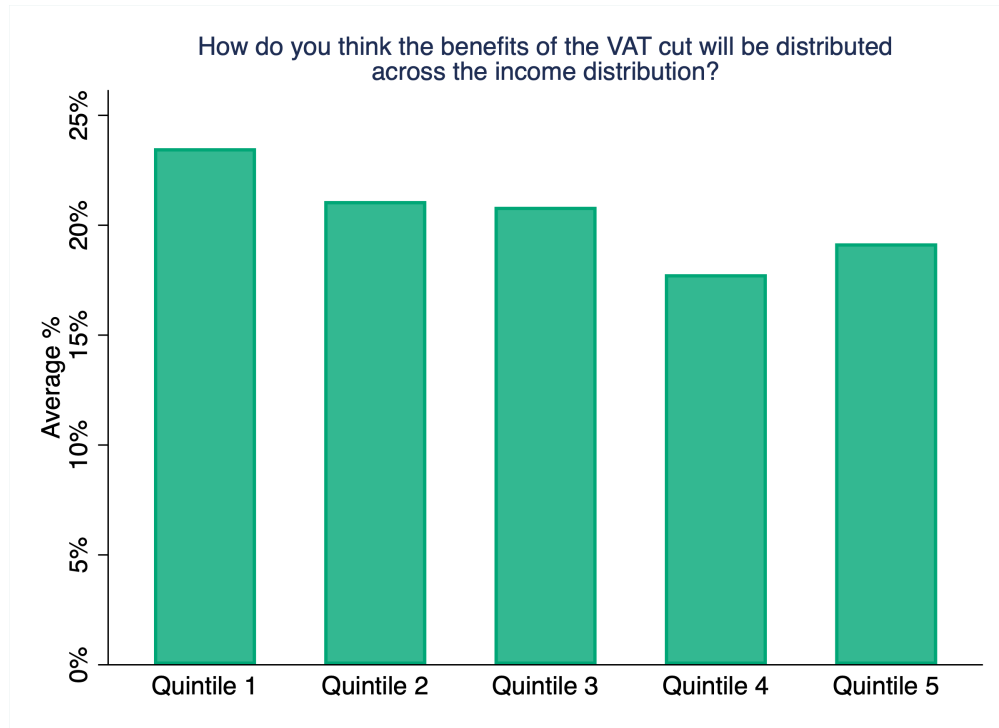


Figure A32: Experts' predictions of incidence of reform by quintile

Note: This figure shows experts' predictions of incidence of reform by quintile.

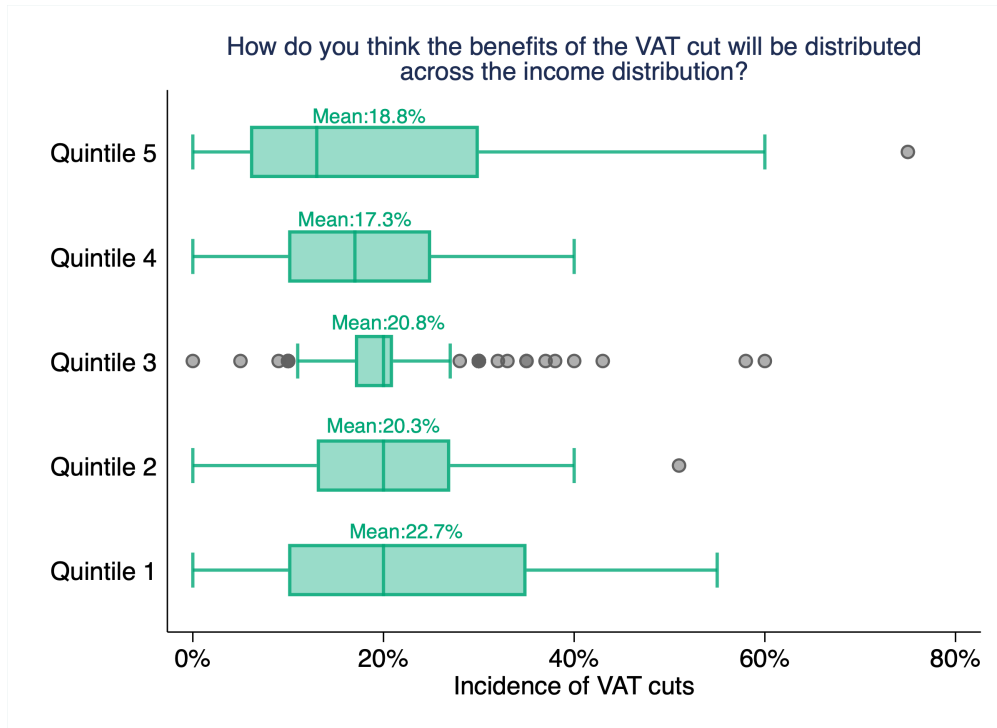


Figure A33: Experts' predictions of incidence of reform by quintile (boxplot)

Note: This figure shows experts' predictions of incidence of reform by quintile (boxplot).

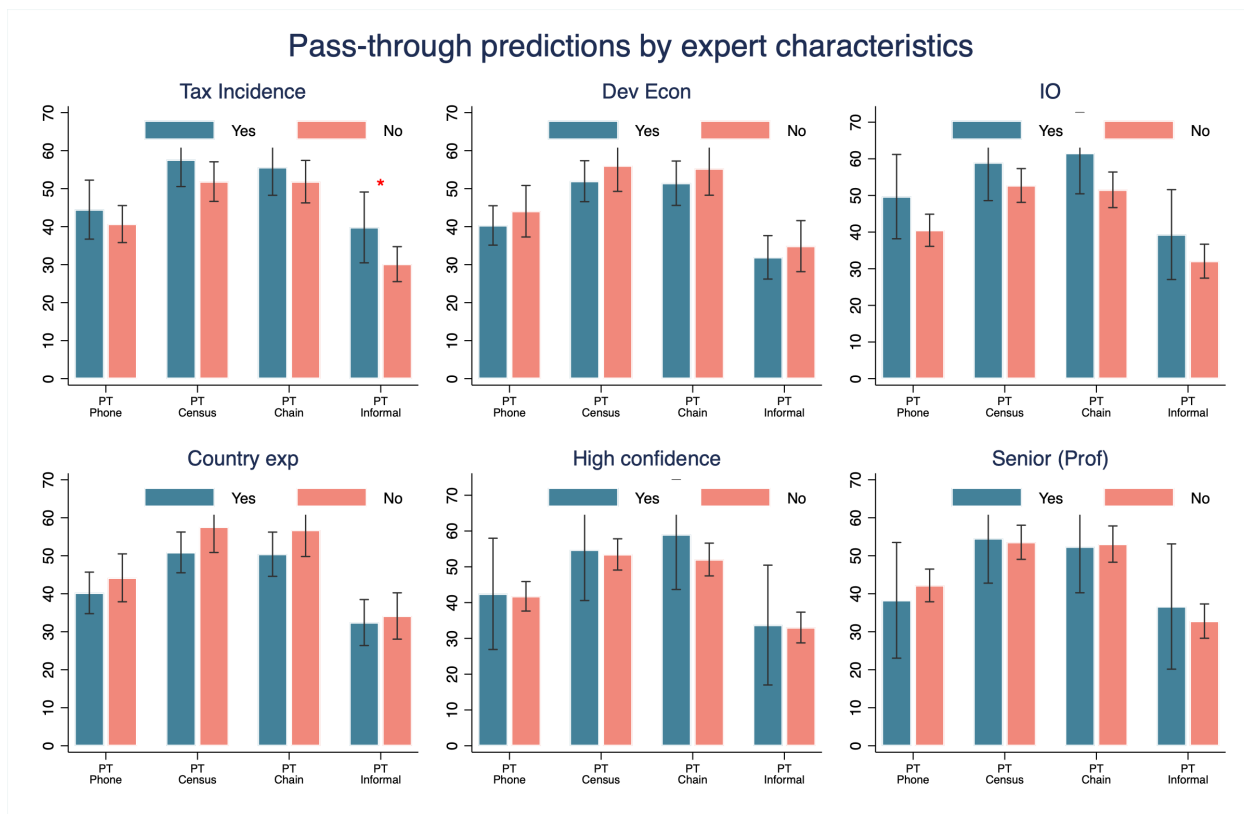


Figure A34: Heterogeneity in experts' predictions of pass-through

Note: This figure shows heterogeneity in experts' pass-through predictions.

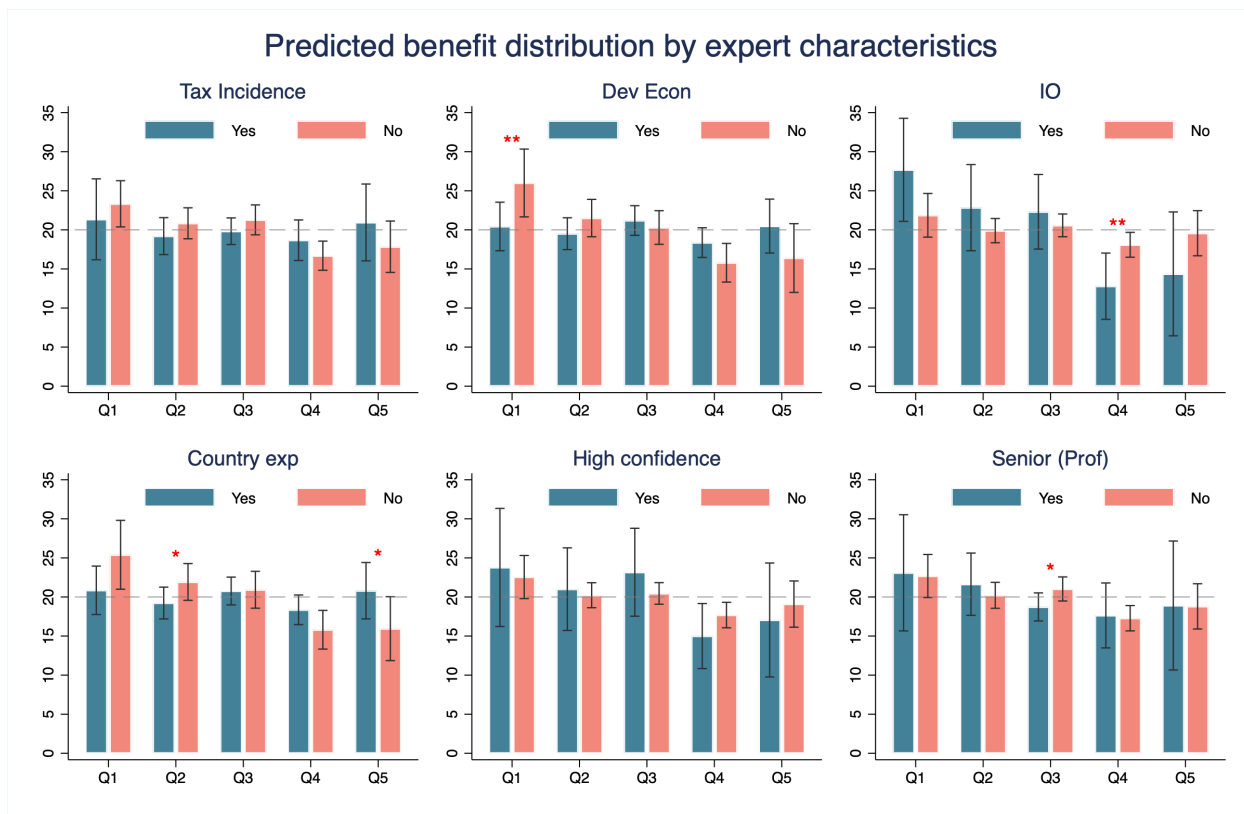


Figure A35: Heterogeneity in experts' predictions of incidence

Note: This figure shows heterogeneity in experts' predictions of incidence.

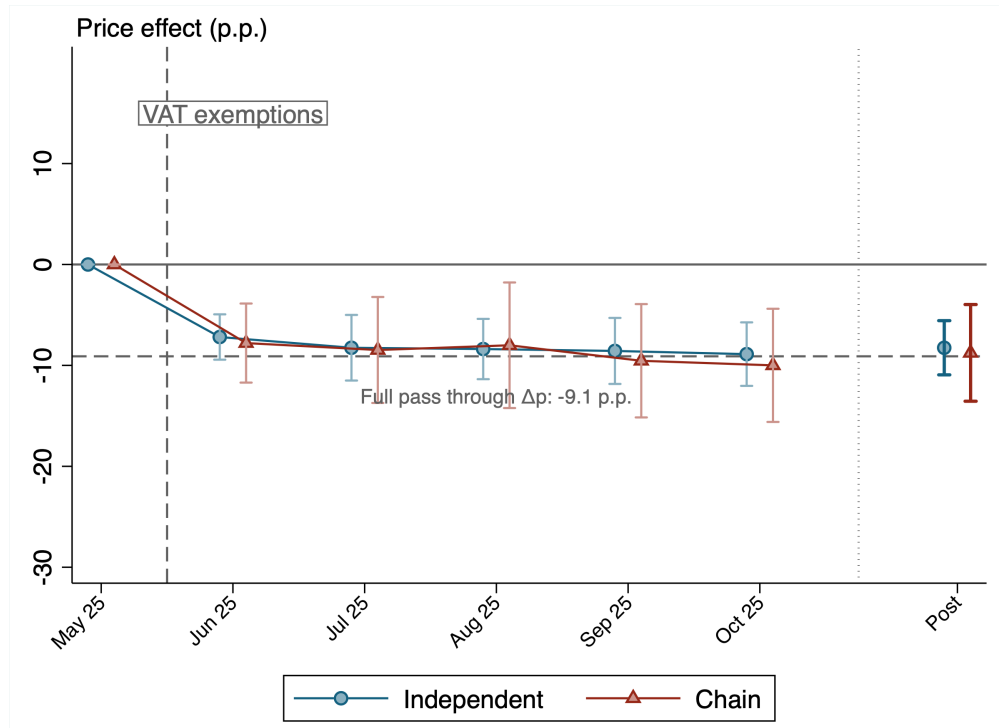


Figure A36: Differences-in-Differences results by chain vs independent stores

Note: This figure presents event study estimates including the average of post-reform event study coefficients (reference = May 2025), by chain vs independent stores (balanced panel).

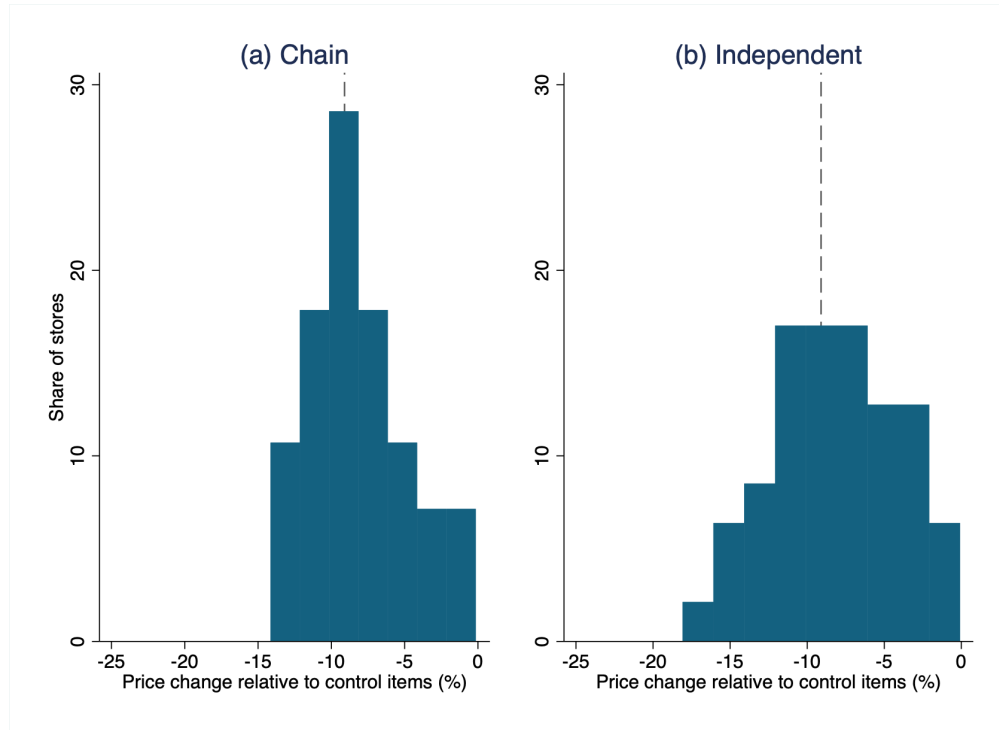


Figure A37: Heterogeneity of pass-through based on chain vs independent stores

Note: This figure shows how the pass-through rate varies by chain vs independent stores, using the supermarket census data.

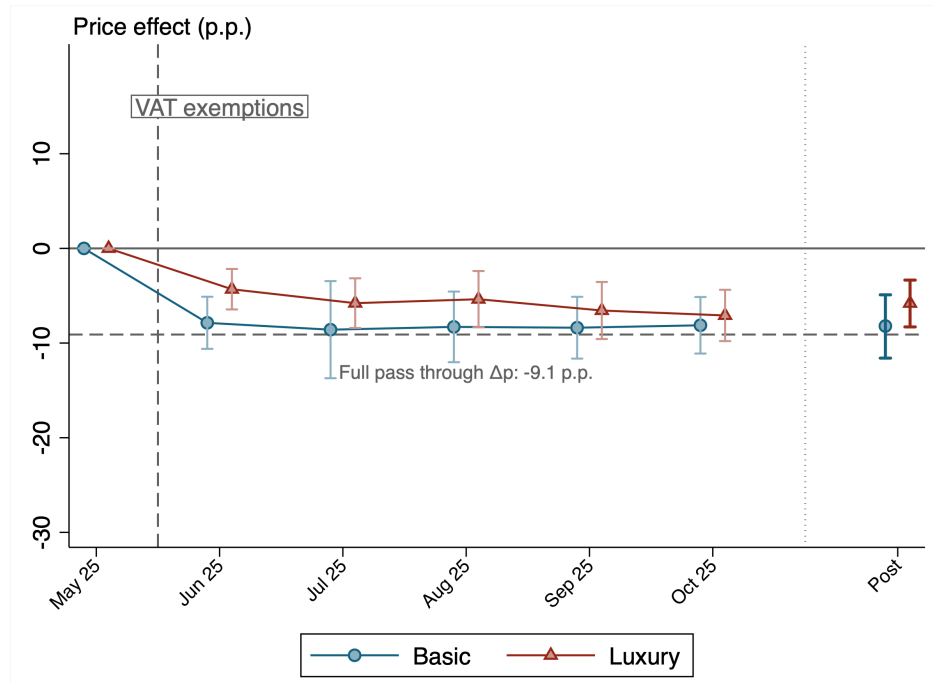


Figure A38: Differences-in-Differences results by product type

Note: This figure presents event study estimates including the average of post-reform event study coefficients (reference = May 2025), by basic vs luxury products (balanced panel).

Appendix C – Expert Prediction Survey

*External asset (not generated by this replication package):
Expertsurvey.pdf – full expert prediction survey instrument*